bond_flip

process design review · 25 steps · generated by CarbonEDA v0 on 2026-11-09 00:56 EST

96.2

kgCO2e per wafer start



12.2
EPW energy · 13%

78.4
MPW materials · 82%

5.5
GPW gases · 6%

wafer 300 mm · yield 1 · carbon intensity 0.43 kgCO2e/kWh · facility overhead 0.4 x tool energy
EPW 12.2 = tool 8.7 + facility overhead 3.5 kgCO2e (20 + 8 kWh)

flow technologies/examples/bond_flip/bond_flip.csv
layout technologies/examples/bond_flip/bond_flip_layout.gds
structure images from the spx run of 2026-10-31 14:56:40

CarbonEDA parameters

config/carbon_config.yaml and materials.yaml as used for every number in this report

EPW energy

(tool + 0.4 x tool facility overhead) x 0.43 kgCO2e/kWh

process area	kWh / step	steps
RIE	1.33	3
Metallization	2	2
Wet_etch	0.28	9
CVD	1.09	8
ArF-i	0.36	3

GPW process gases

kg x (1 - utilization) x (1 - abatement) x GWP, plus byproducts kg/kg x (1 - abatement) x GWP; defaults: utilization 1.0, abatement 0.9

area	gas	g / step	upstream*	utilization	abatement	byproducts (kg per kg gas)
CVD	NF3	2.54	12.5	0.8	0.95	CF4 0.046, C2F6 0.045, CHF3 0.025, CH2F2 0.00086,...
RIE	C4F8	0.16	12.1	0.82	0.98	CF4 0.045, C2F6 0.031, C4F6 0.018, CHF3 0.012, CH...
RIE	SF6	0.439	125.6	0.71	0.95	CF4 0.04, C2F6 0.044, CHF3 0.0037, CH2F2 2.9e-05

* kgCO2e per kg of gas made, counted in MPW

GWP (kgCO2e per kg, 100-yr) and abatement of every gas emitted

gas	GWP	abatement
CF4	7390	0.89
C2F6	12200	0.95
C4F6	0	0.98
C4F8	10300	0.98
CHF3	14800	0.98

gas	GWP	abatement
CH2F2	675	0.98
CH3F	116	0.98
NF3	17200	0.95
SF6	22800	0.95

MPW materials

blanket film over a 300 mm wafer: thickness x area x density x GWP; the wafer itself 26000 gCO2e (300 mm); yield 1

material	g/cm³	gCO2e/g
Cu	8.96	6.09

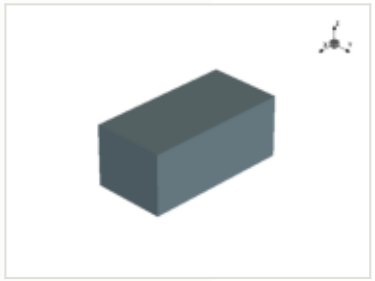
substrate

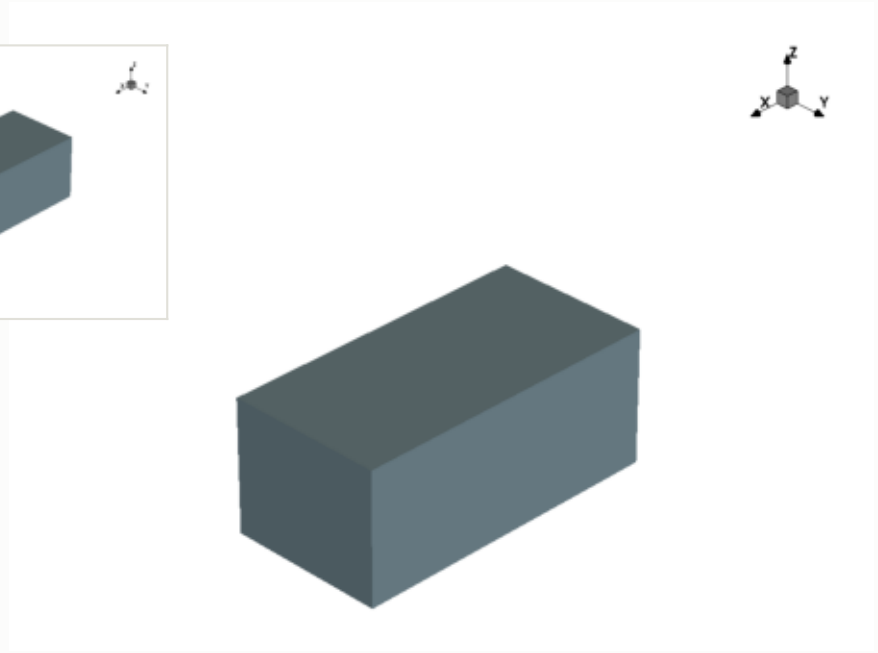
substrate · Wafer

26	52	27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





this step

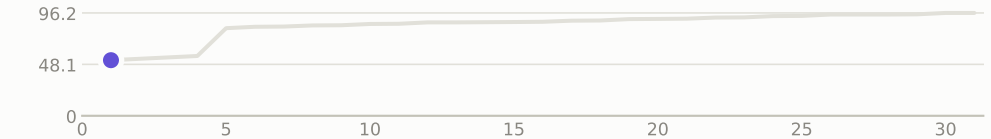
process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	26 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**26 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

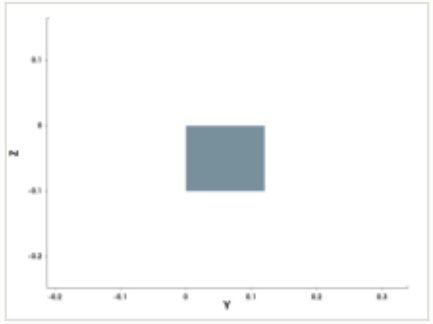
carbon so far, of the whole flow (kgCO2e)



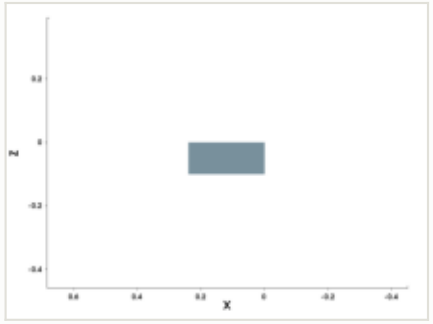
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



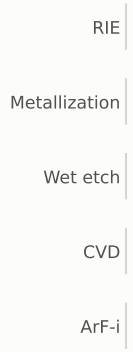
y cut at 0.060 µm



mask for this step

no mask for this step

steps by process area



materials deposited (g)



gases emitted (g)

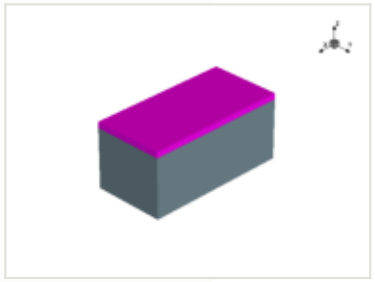
SiGe etch stop

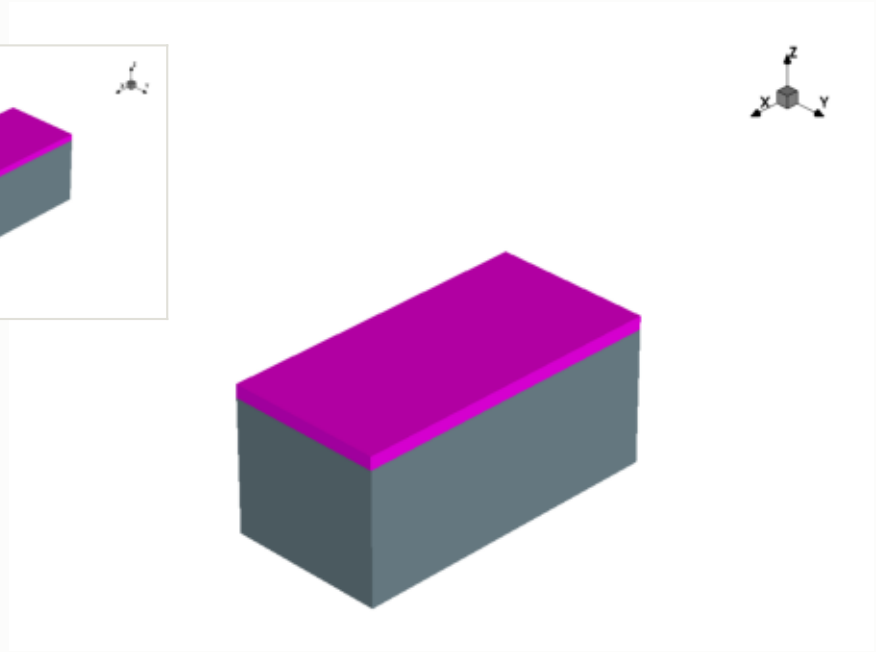
depo_anisotropic · CVD

1.31	53.3	1.4%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

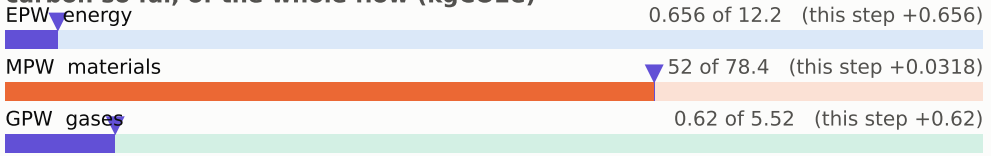




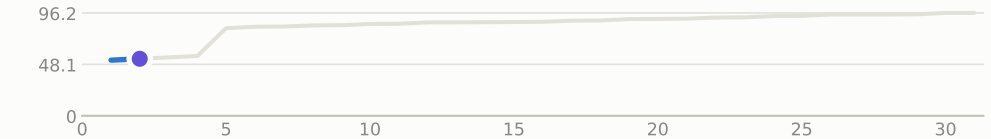
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.01 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

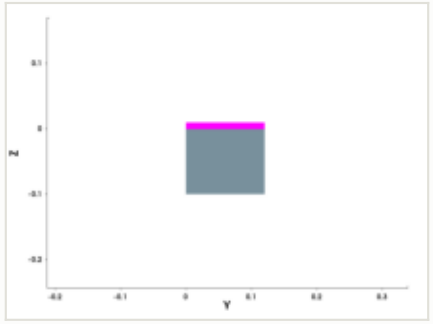
carbon so far, of the whole flow (kgCO2e)



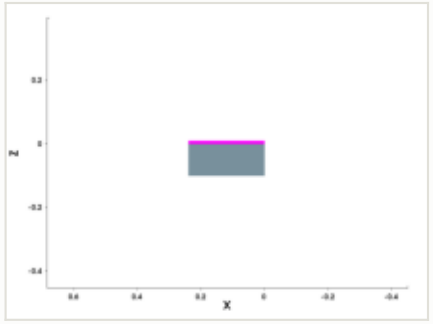
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



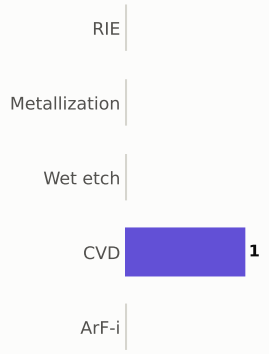
y cut at 0.060 µm



mask for this step

no mask for this step

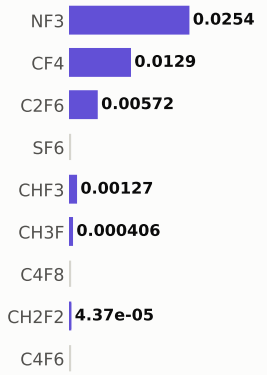
steps by process area



materials deposited (g)



gases emitted (g)



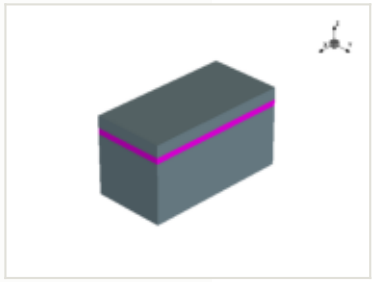
Si channel layer

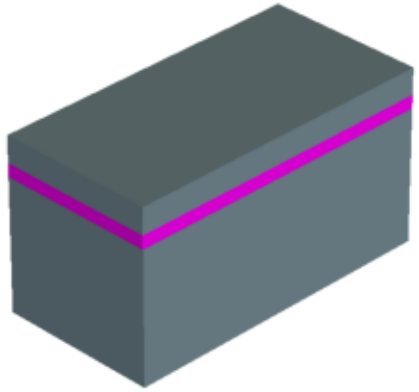
depo_anisotropic · CVD

1.31	54.6	1.4%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



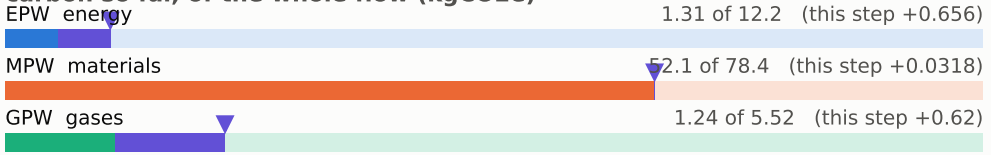


this step

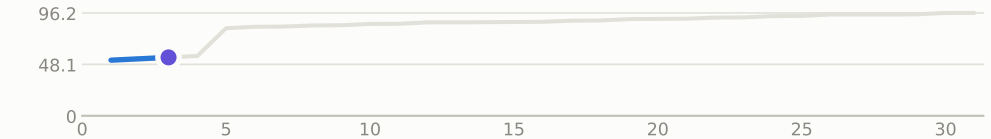
process area	CVD
material	—
thickness	0.02 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

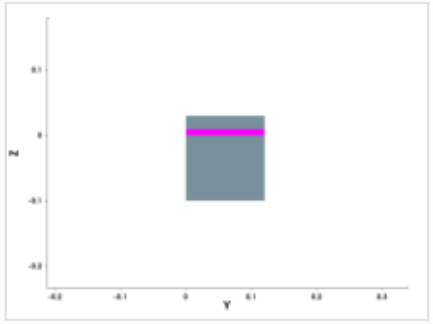
carbon so far, of the whole flow (kgCO2e)



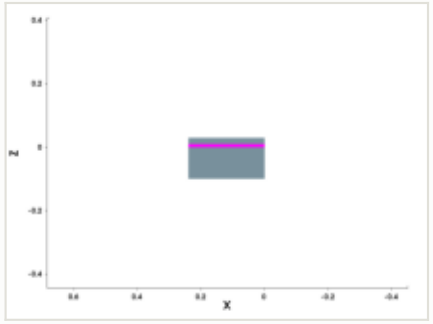
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



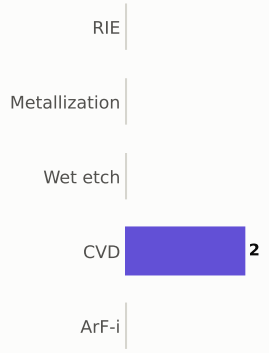
y cut at 0.060 µm



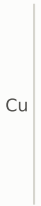
mask for this step

no mask for this step

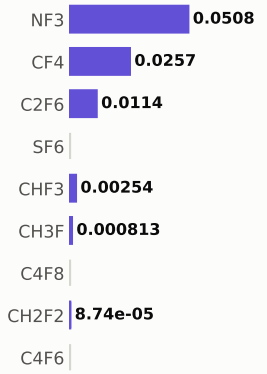
steps by process area



materials deposited (g)



gases emitted (g)



Bonding oxide

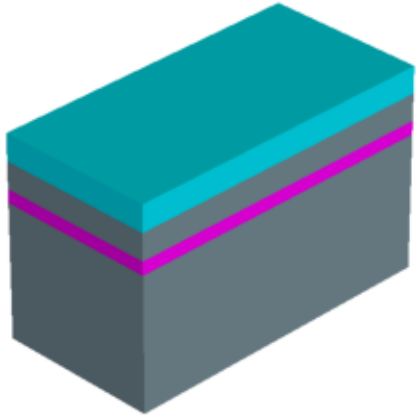
depo_anisotropic · CVD

1.31	55.9	1.4%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

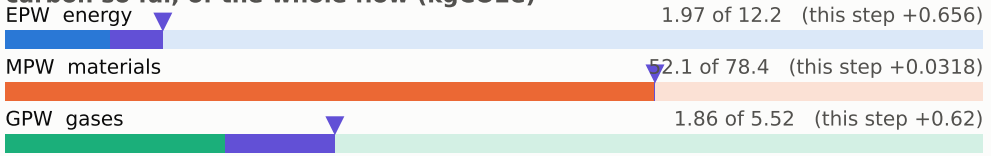




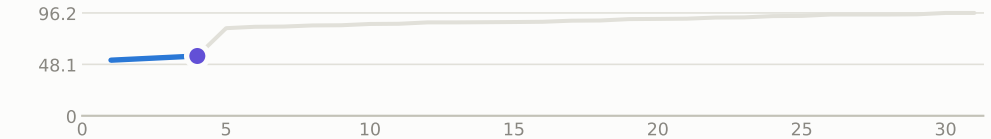
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.02 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

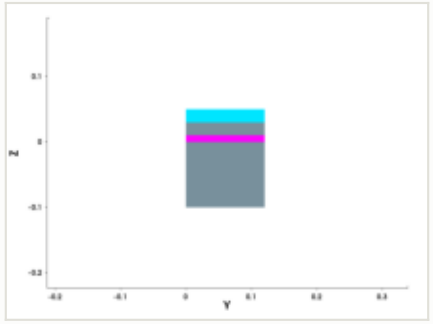
carbon so far, of the whole flow (kgCO2e)



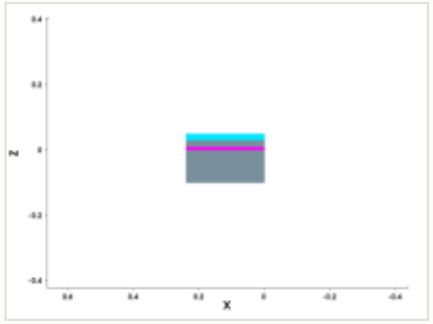
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



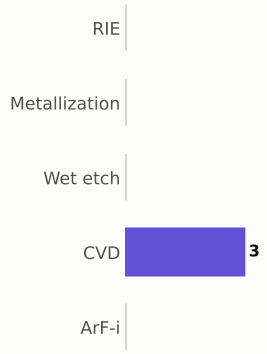
y cut at 0.060 µm



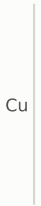
mask for this step

no mask for this step

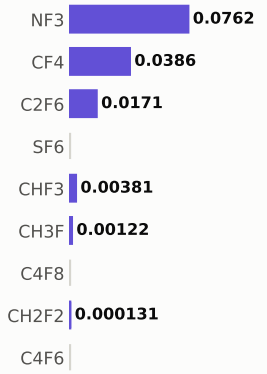
steps by process area



materials deposited (g)



gases emitted (g)



substrate

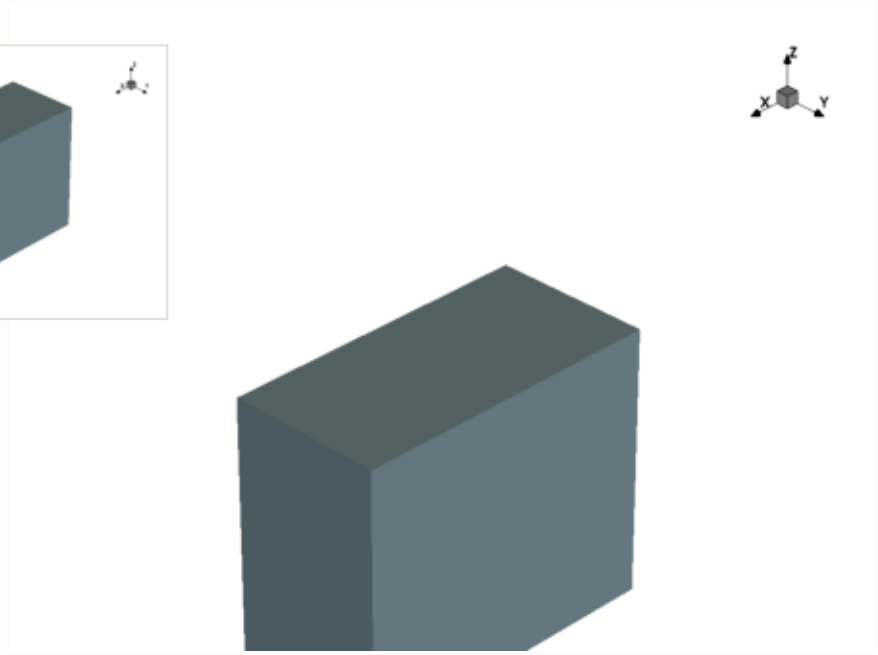
substrate · Wafer

26	81.9	27%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



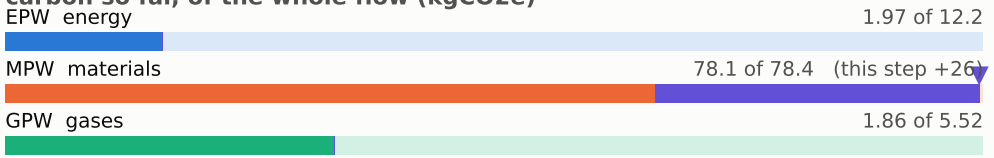


this step

process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	26 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**26 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

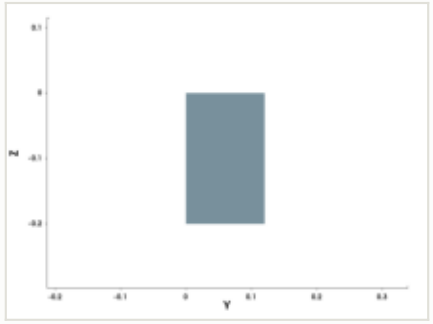
carbon so far, of the whole flow (kgCO2e)



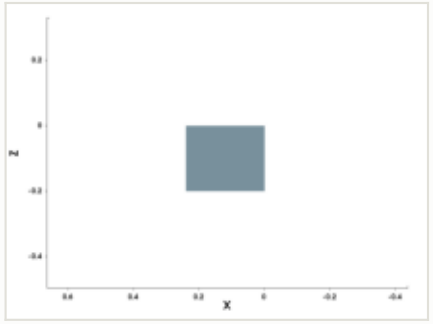
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



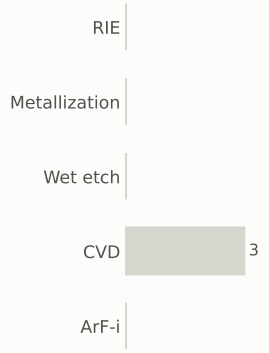
y cut at 0.060 µm



mask for this step

no mask for this step

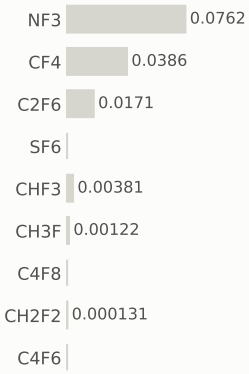
steps by process area



materials deposited (g)



gases emitted (g)



ILD

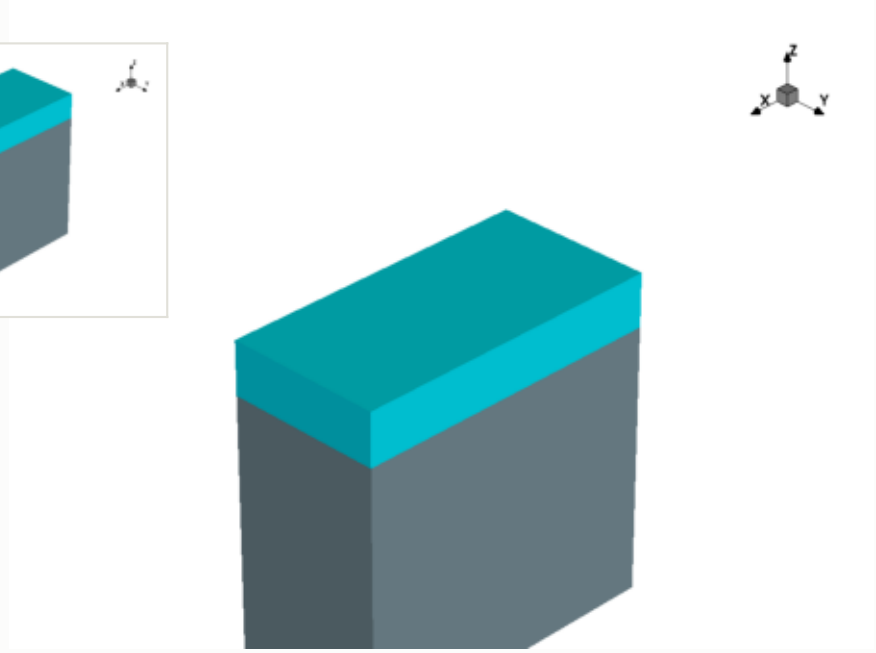
depo_anisotropic · CVD

1.31	83.2	1.4%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



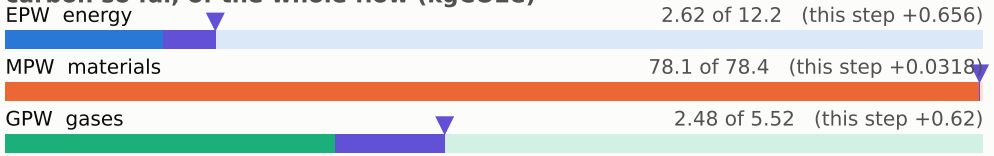


this step

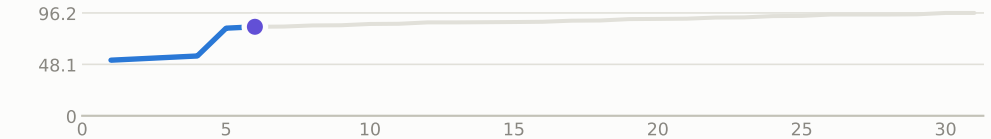
process area	CVD
material	—
thickness	0.04 µm
material GWP	—
input gases	NF3 2.54 g
energy	1.09 kWh + 0.436 overhead

EPW	0.656 kgCO2e
EPW tool	0.469 kgCO2e
EPW overhead	0.187 kgCO2e
MPW	0.0318 kgCO2e (gas 0.0318)
GPW	0.62 kgCO2e
embodied carbon	**1.31 kgCO2e**

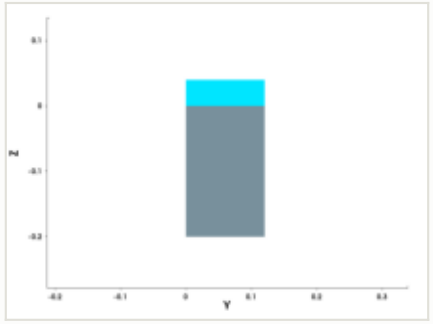
carbon so far, of the whole flow (kgCO2e)



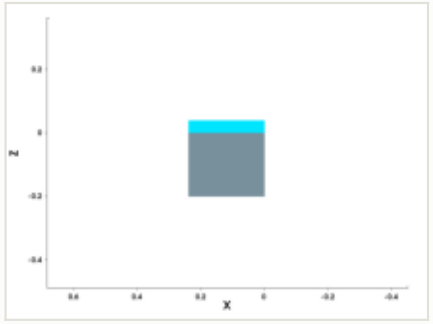
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



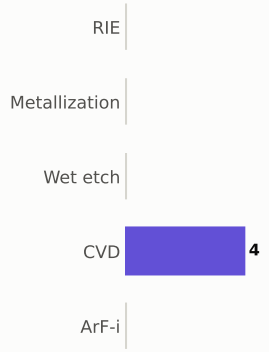
y cut at 0.060 µm



mask for this step

no mask for this step

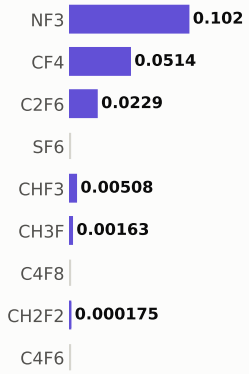
steps by process area



materials deposited (g)



gases emitted (g)



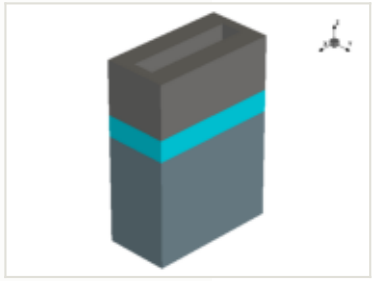
Bottom metal litho

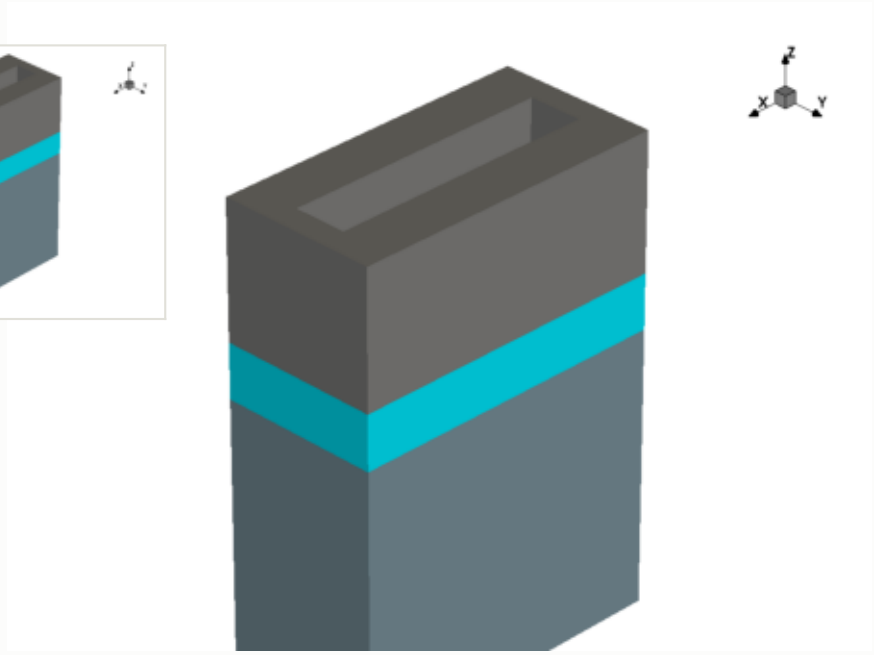
litho_patterning · ArF-i

0.217	83.5	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

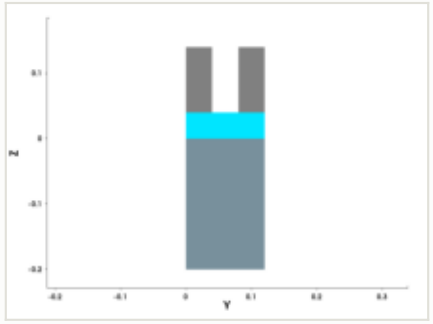
3D structure after this step

zoomed to fit

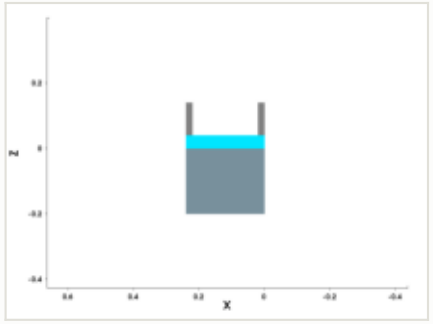




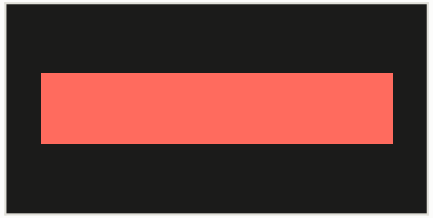
x cut at 0.120 µm



y cut at 0.060 µm



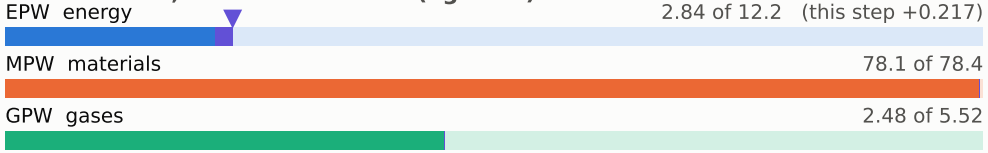
mask: layer 1:0 (GDSII)



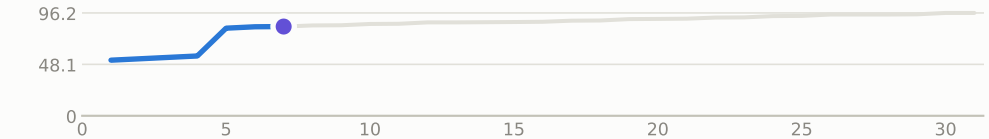
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

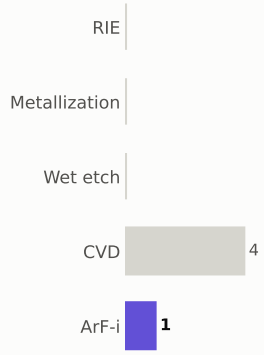
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)



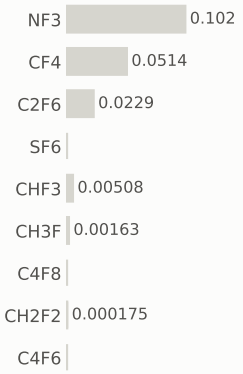
steps by process area



materials deposited (g)



gases emitted (g)



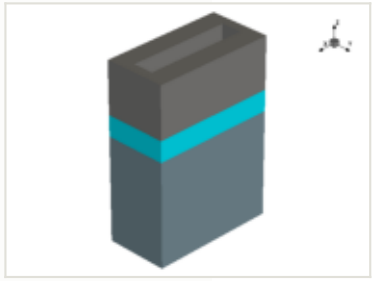
Bottom metal trench etch

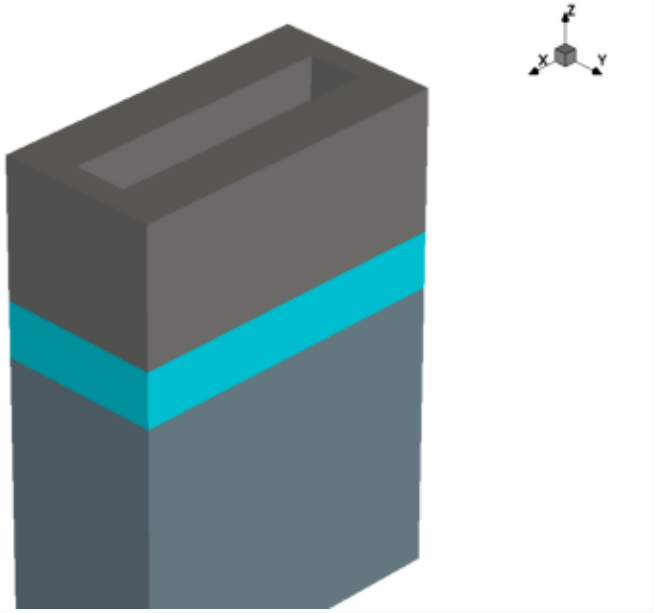
etch_anisotropic · RIE

1.04	84.5	1.1%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

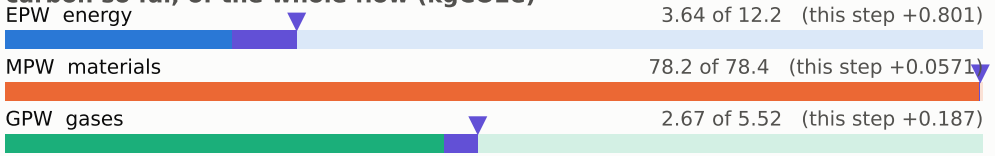




this step

process area	RIE	EPW	0.801 kgCO2e
material	—	EPW tool	0.572 kgCO2e
thickness	—	EPW overhead	0.229 kgCO2e
material GWP	—	MPW	0.0571 kgCO2e (gas 0.0571)
input gases	C4F8 0.16 g, SF6 0.439 g	GPW	0.187 kgCO2e
energy	1.33 kWh + 0.532 overhead	embodied carbon	**1.04 kgCO2e**

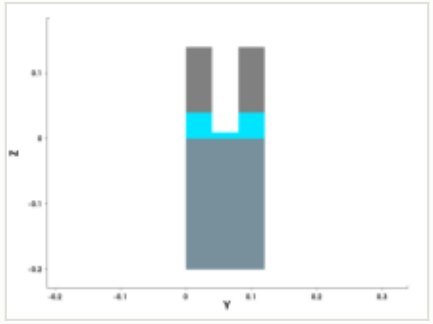
carbon so far, of the whole flow (kgCO2e)



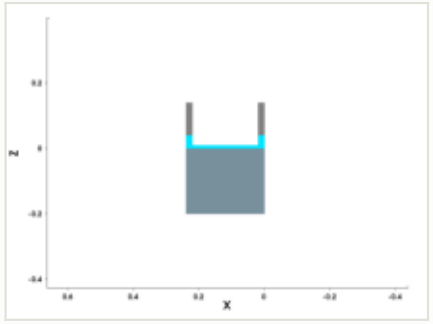
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



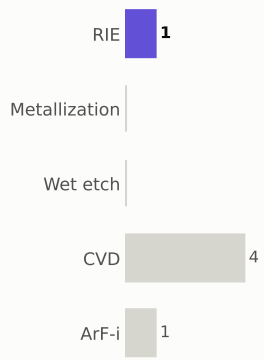
y cut at 0.060 µm



mask for this step

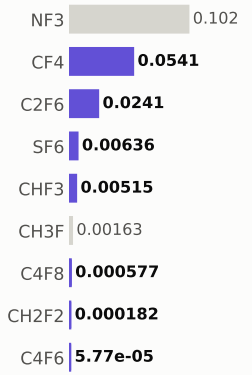
no mask for this step

steps by process area



materials deposited (g)

gases emitted (g)



Resist strip

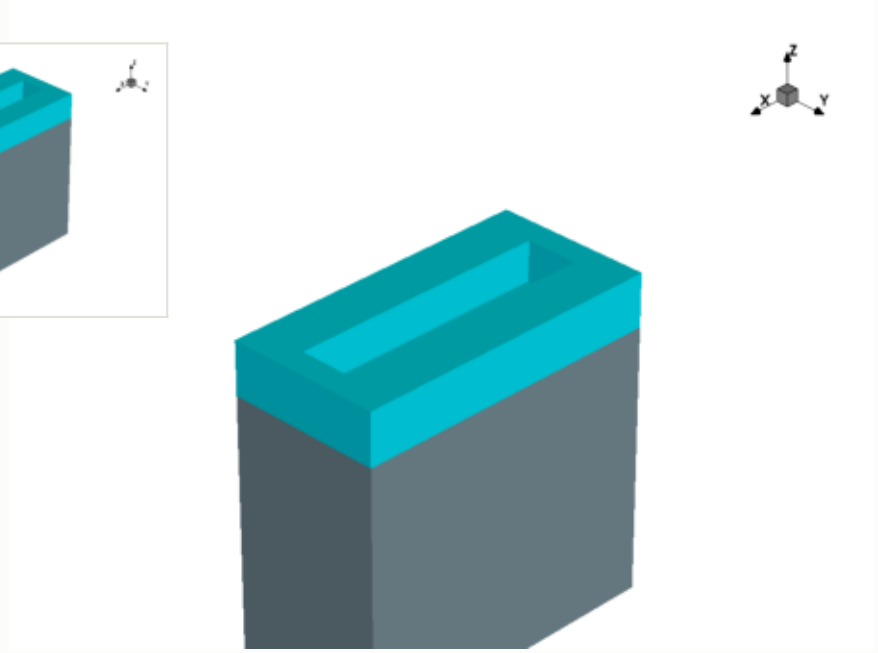
etch_strip · Wet_etch

0.169	84.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

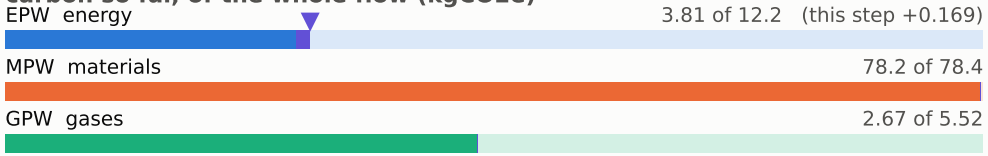




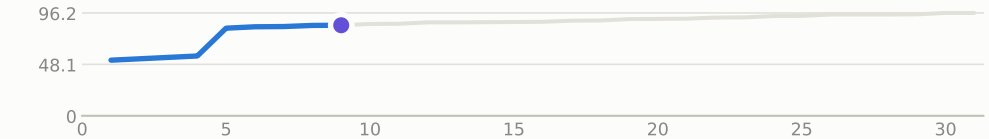
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

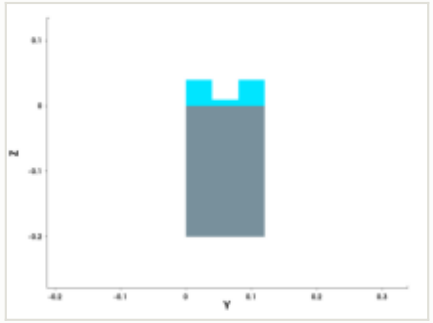
carbon so far, of the whole flow (kgCO2e)



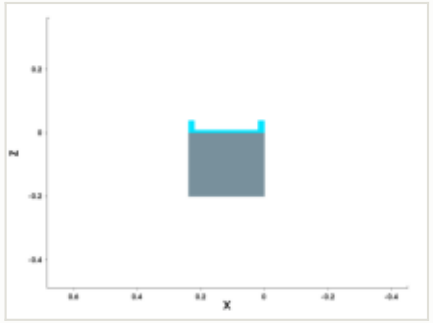
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



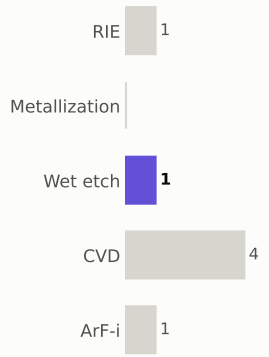
y cut at 0.060 µm



mask for this step

no mask for this step

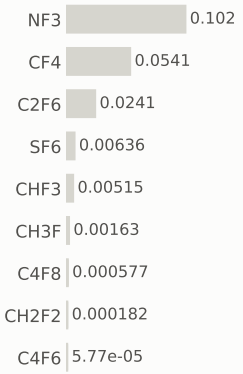
steps by process area



materials deposited (g)



gases emitted (g)



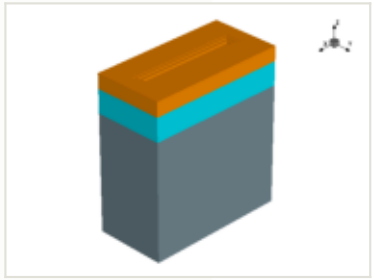
Cu fill

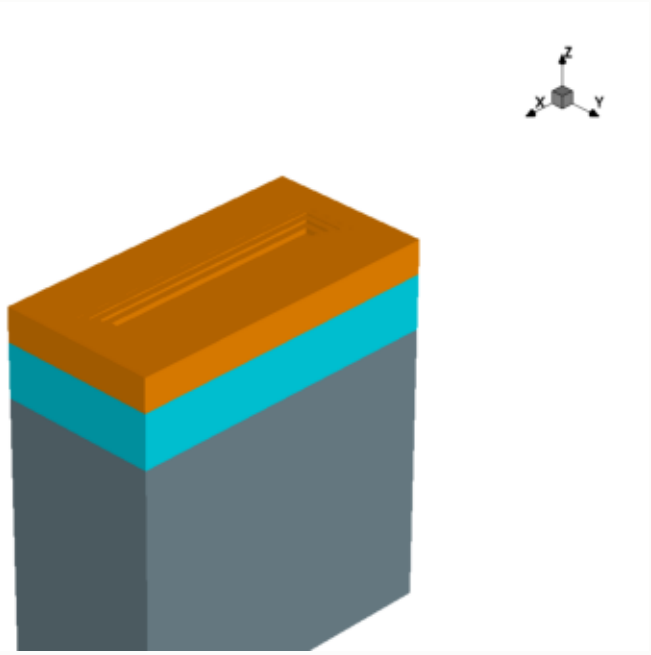
depo_isotropic · Metallization

1.2	85.9	1.3%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



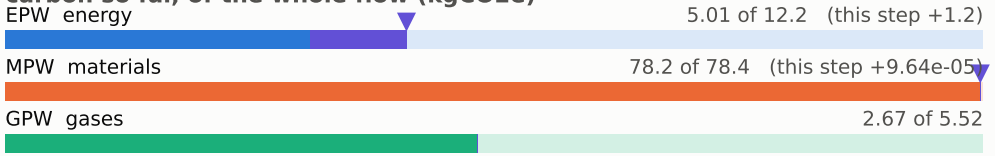


this step

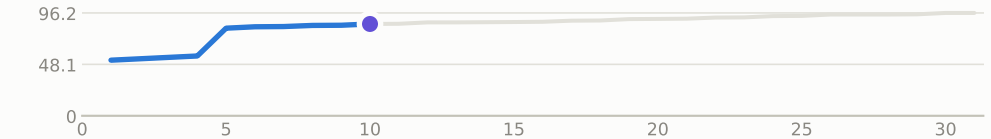
process area	Metallization
material	Cu
thickness	0.025 µm
material GWP	6.09011 gCO2e/g
input gases	—
energy	2 kWh + 0.8 overhead

EPW	1.2 kgCO2e
EPW tool	0.86 kgCO2e
EPW overhead	0.344 kgCO2e
MPW	9.64e-05 kgCO2e
GPW	0 kgCO2e
embodied carbon	**1.2 kgCO2e**

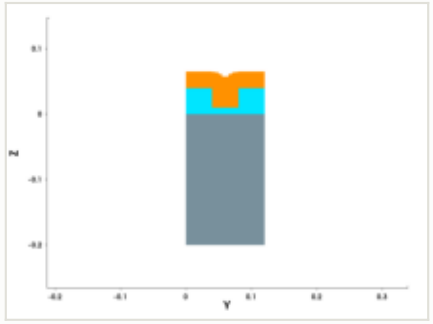
carbon so far, of the whole flow (kgCO2e)



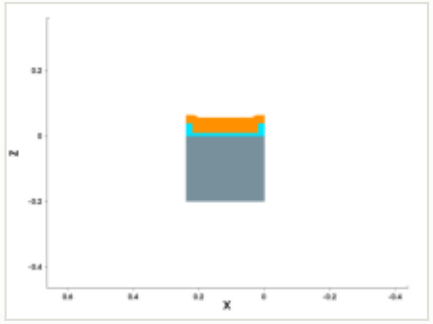
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



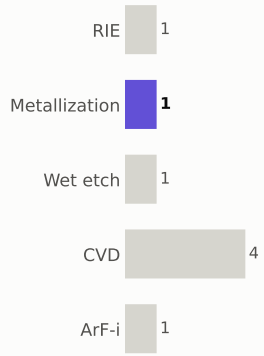
y cut at 0.060 µm



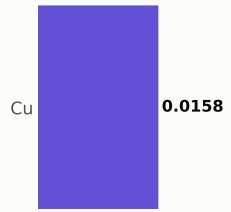
mask for this step

no mask for this step

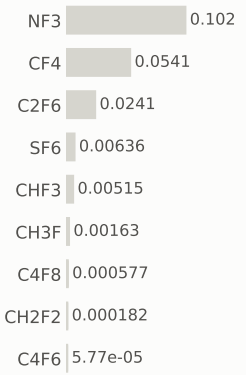
steps by process area



materials deposited (g)



gases emitted (g)



Cu CMP

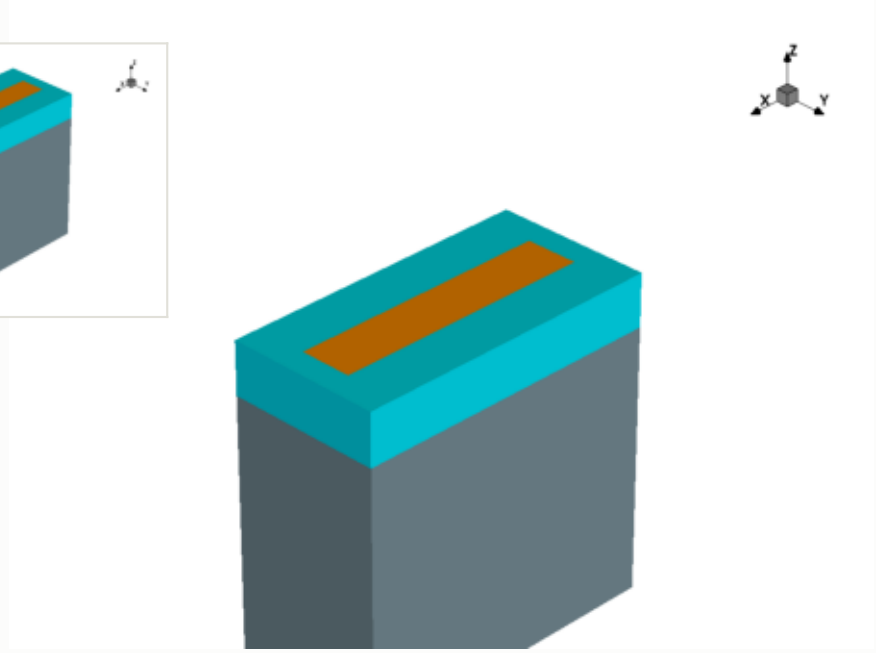
etch_cmp · Wet_etch

0.169	86	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

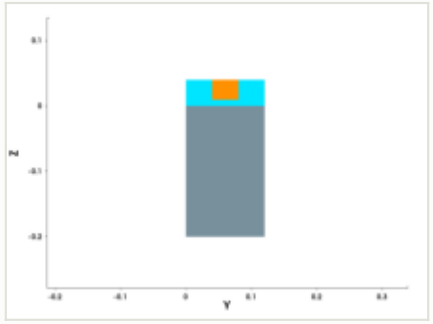
3D structure after this step

zoomed to fit

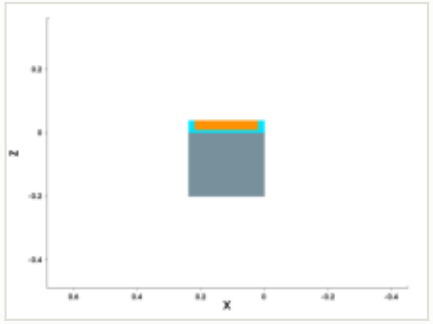




x cut at 0.120 µm



y cut at 0.060 µm



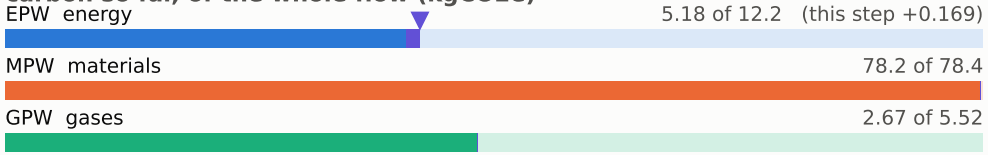
mask for this step

no mask for this step

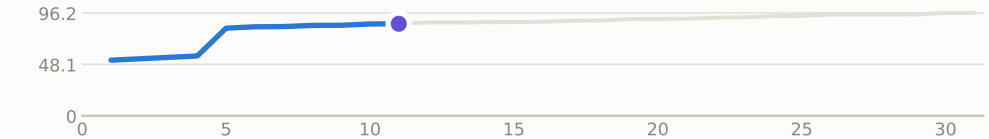
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

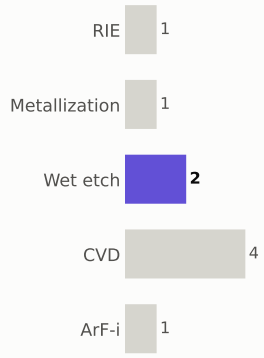
carbon so far, of the whole flow (kgCO2e)



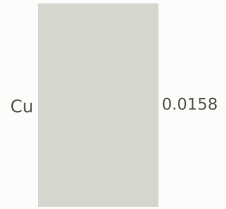
embodied carbon up to each step (kgCO2e per wafer)



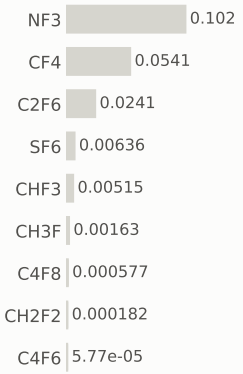
steps by process area



materials deposited (g)



gases emitted (g)



Bonding oxide

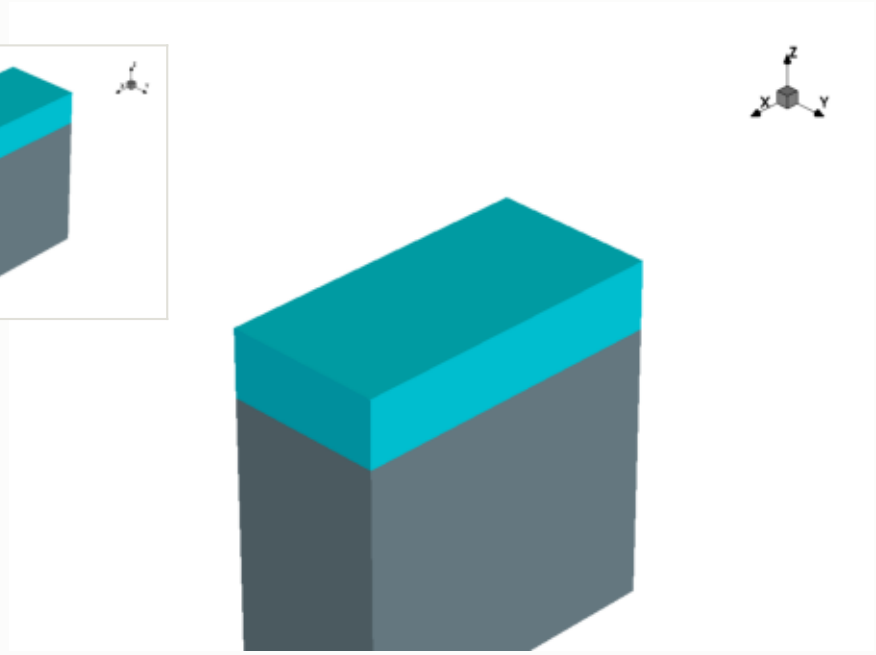
depo_anisotropic · CVD

1.31	87.3	1.4%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

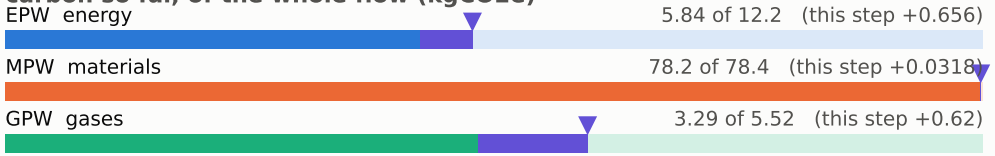




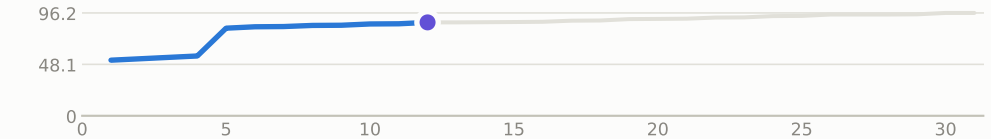
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.01 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

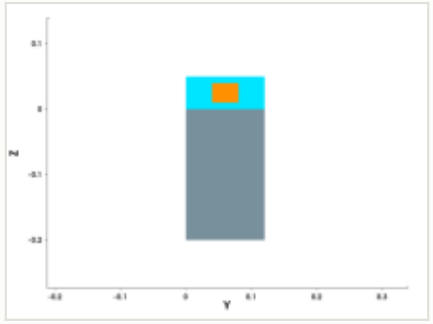
carbon so far, of the whole flow (kgCO2e)



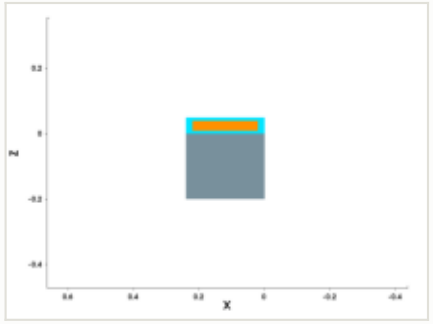
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



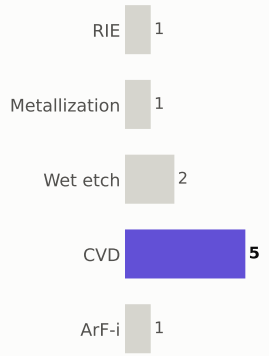
y cut at 0.060 µm



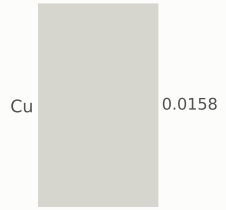
mask for this step

no mask for this step

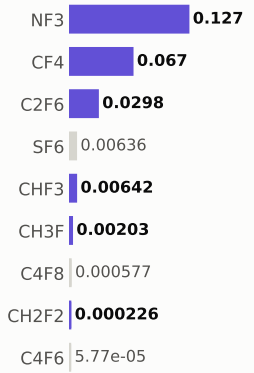
steps by process area



materials deposited (g)



gases emitted (g)



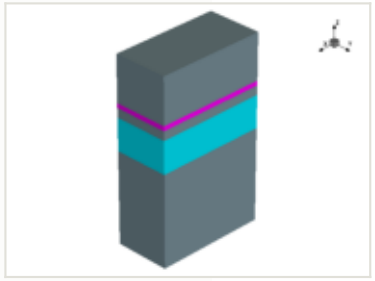
Bond donor wafer

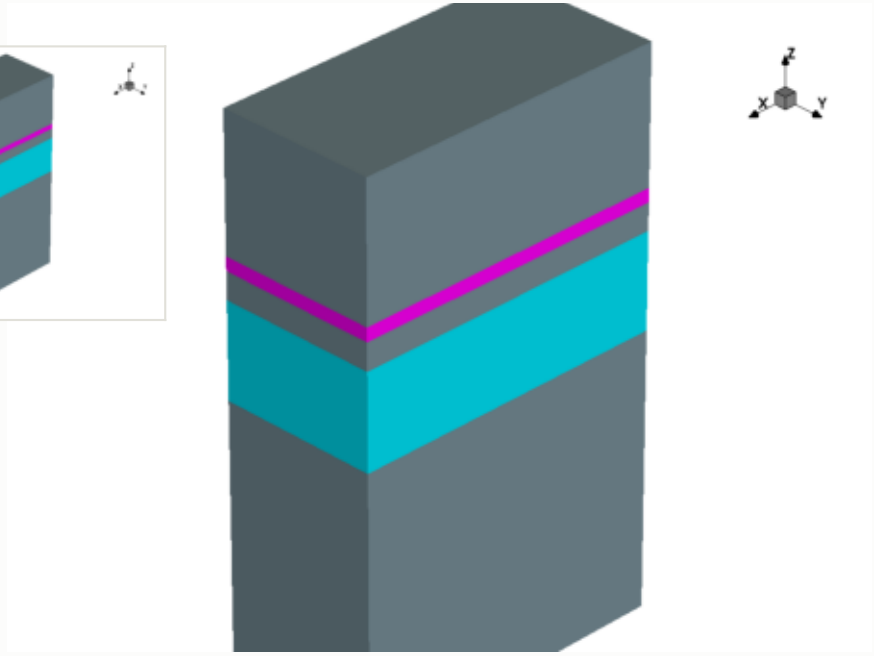
bond_structure · Wafer

0	87.3	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit





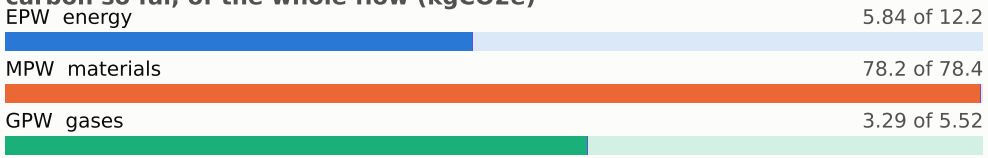
this step

process area	Wafer
material	—
thickness	—
material GWP	—
input gases	—
energy	0 kWh + 0 overhead

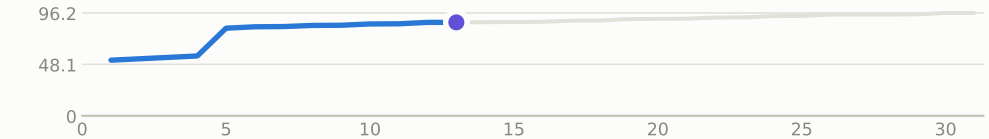
EPW	0 kgCO2e
EPW tool	0 kgCO2e
EPW overhead	0 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

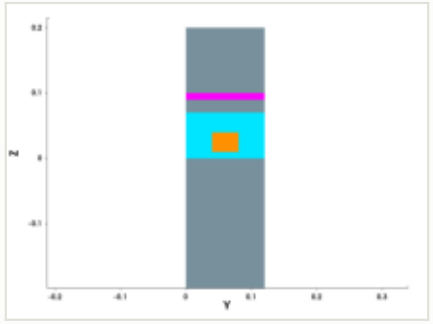
carbon so far, of the whole flow (kgCO2e)



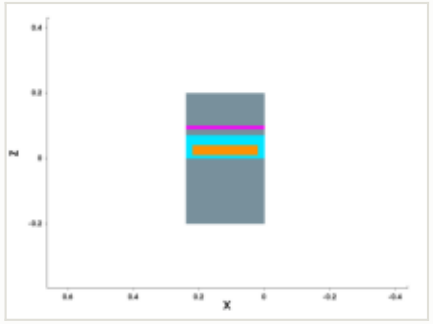
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



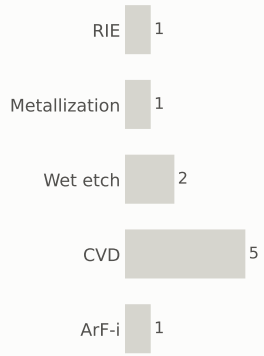
y cut at 0.060 µm



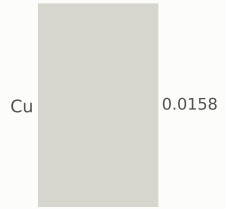
mask for this step

no mask for this step

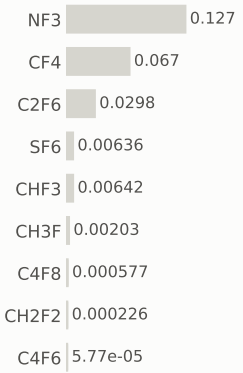
steps by process area



materials deposited (g)



gases emitted (g)



Donor substrate CMP

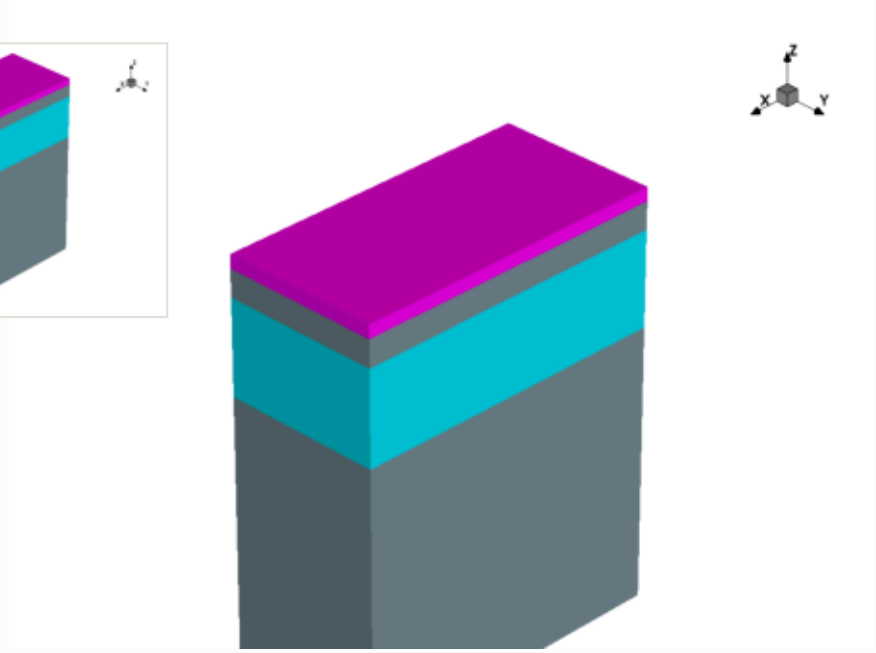
etch_cmp · Wet_etch

0.169	87.5	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



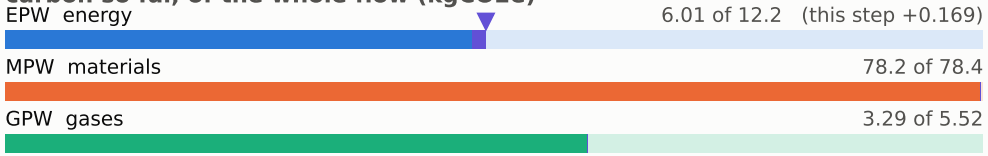


this step

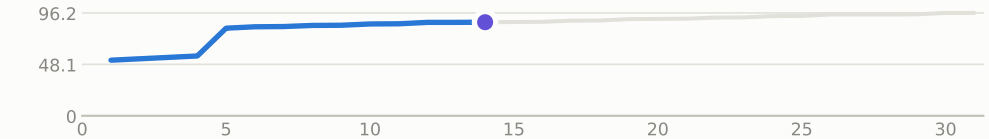
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

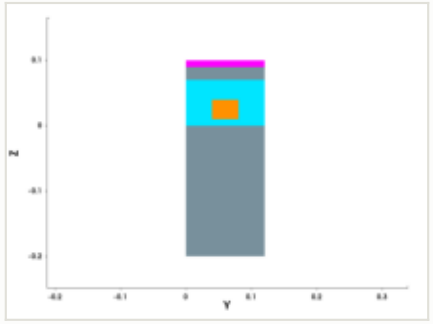
carbon so far, of the whole flow (kgCO2e)



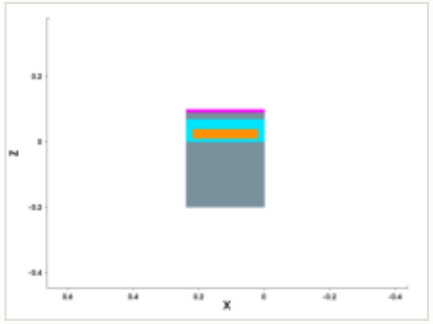
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



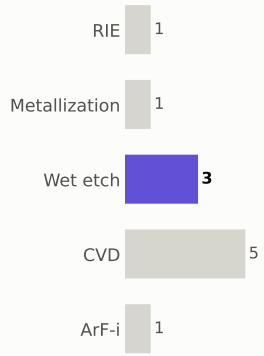
y cut at 0.060 µm



mask for this step

no mask for this step

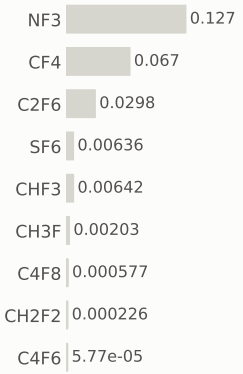
steps by process area



materials deposited (g)



gases emitted (g)



Etch stop removal

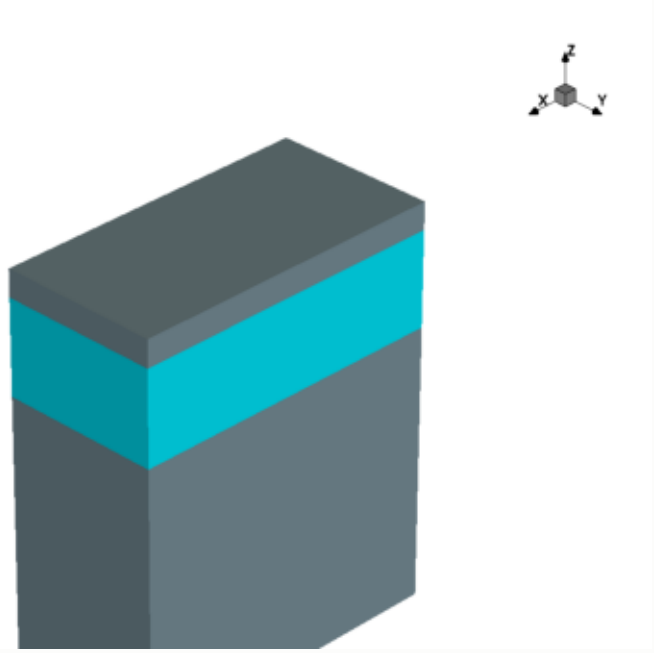
etch_cmp · Wet_etch

0.169	87.7	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



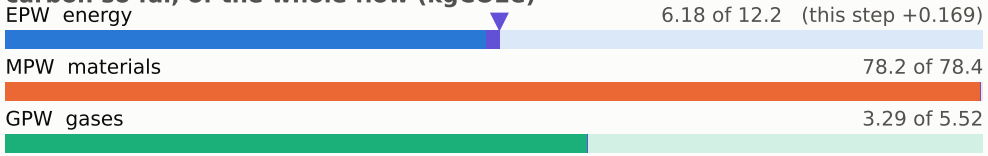


this step

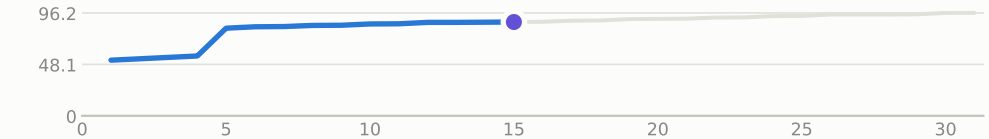
process area	Wet_etch
material	—
thickness	—
material GWP	—
input gases	—
energy	0.28 kWh + 0.112 overhead

EPW	0.169 kgCO2e
EPW tool	0.12 kgCO2e
EPW overhead	0.0482 kgCO2e
MPW	0 kgCO2e
GPW	0 kgCO2e
embodied carbon	**0.169 kgCO2e**

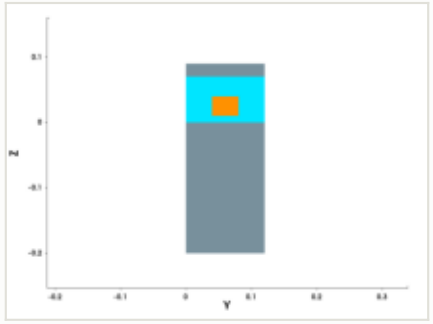
carbon so far, of the whole flow (kgCO2e)



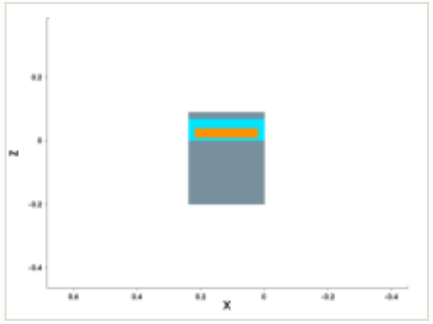
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



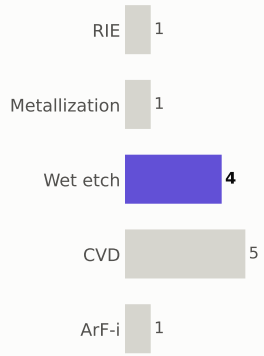
y cut at 0.060 µm



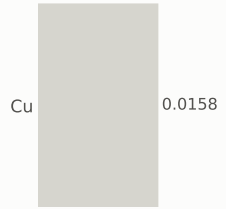
mask for this step

no mask for this step

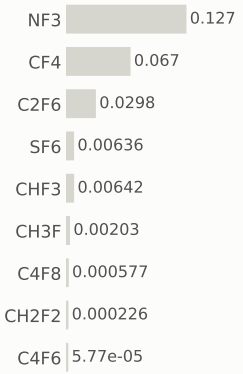
steps by process area



materials deposited (g)



gases emitted (g)



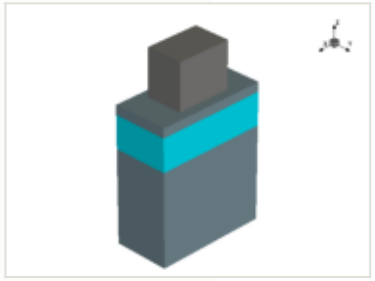
Island litho

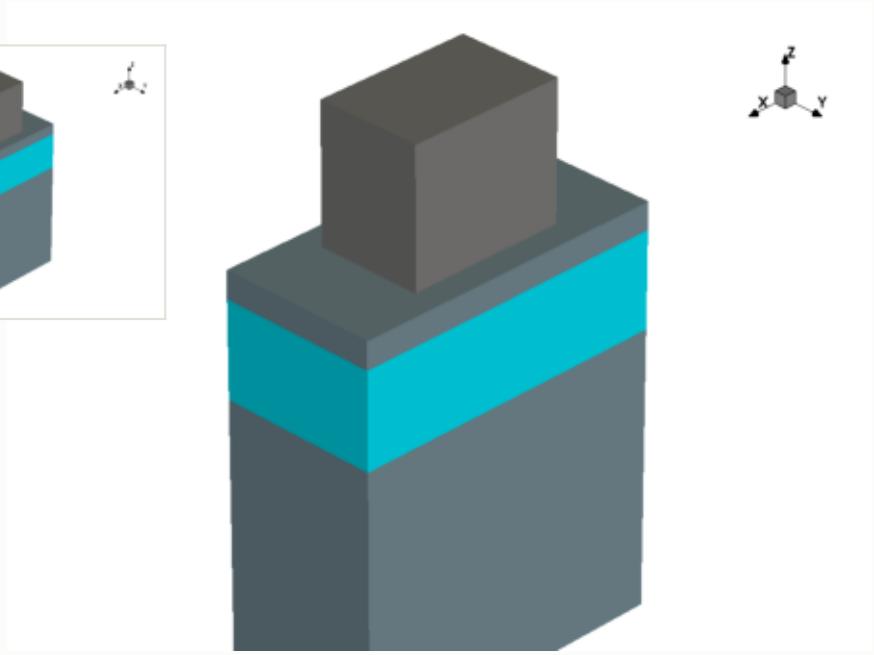
litho_patterning · ArF-i

0.217	87.9	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

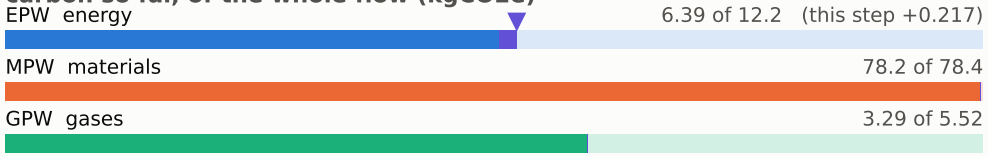




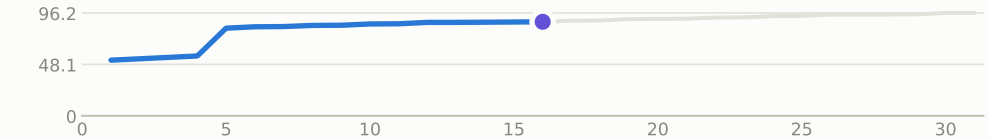
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

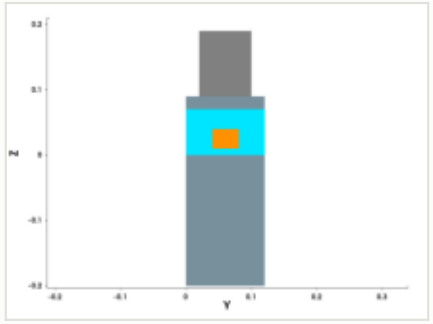
carbon so far, of the whole flow (kgCO2e)



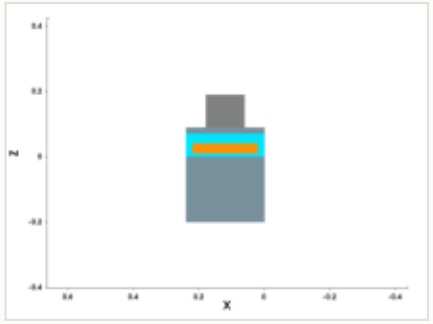
embodied carbon up to each step (kgCO2e per wafer)



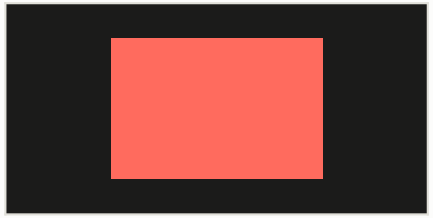
x cut at 0.120 µm



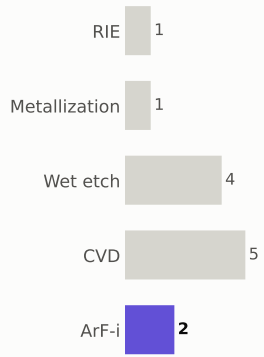
y cut at 0.060 µm



mask: layer 2:0, positive (GDSII)



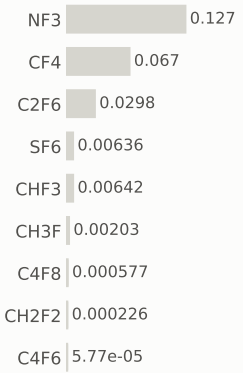
steps by process area



materials deposited (g)



gases emitted (g)



Island etch

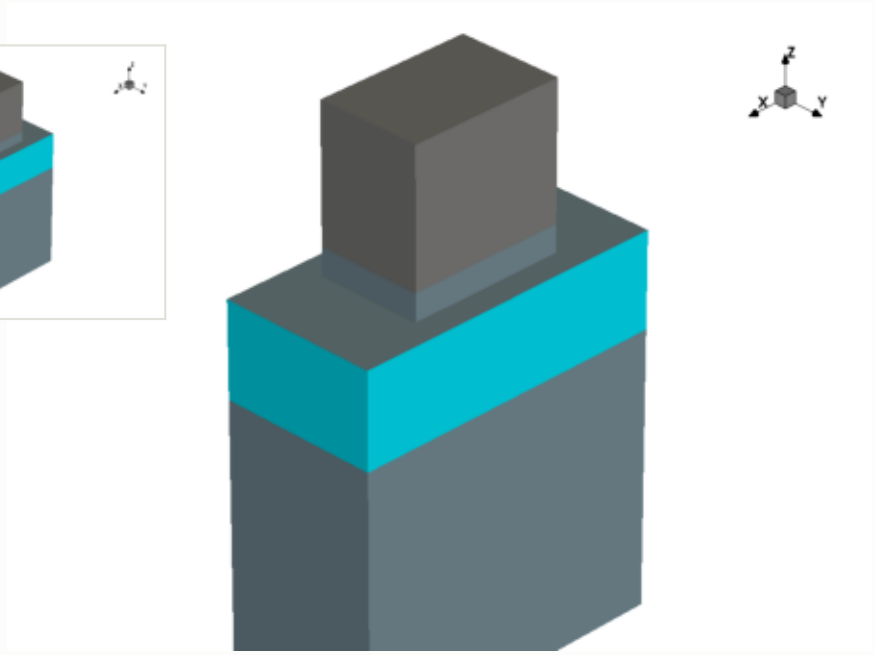
etch_anisotropic · RIE

1.04	88.9	1.1%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



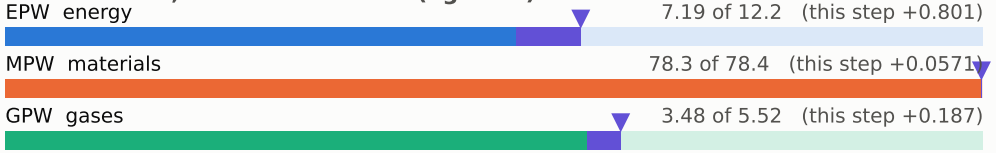


this step

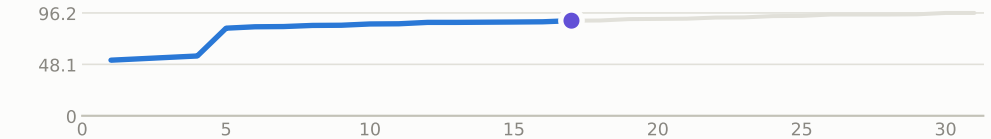
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

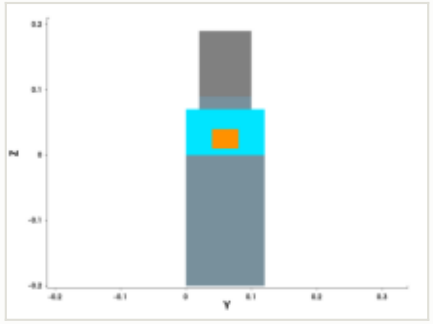
carbon so far, of the whole flow (kgCO2e)



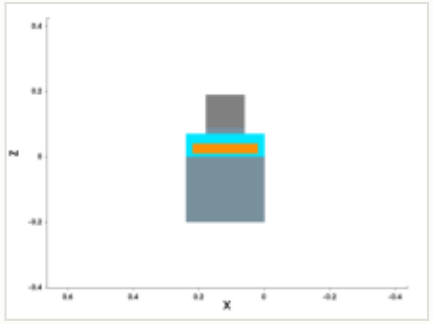
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



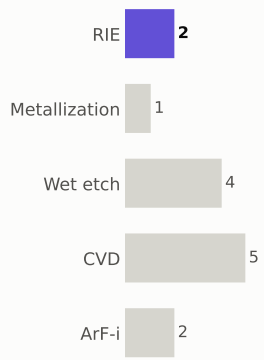
y cut at 0.060 µm



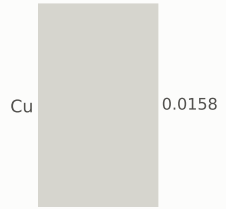
mask for this step

no mask for this step

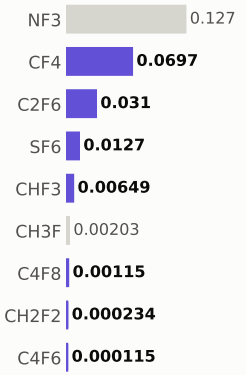
steps by process area



materials deposited (g)



gases emitted (g)



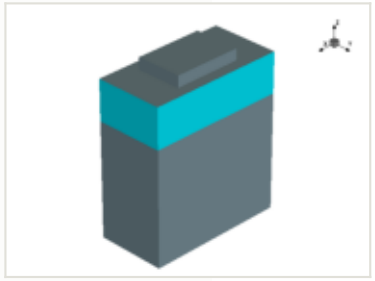
Resist strip

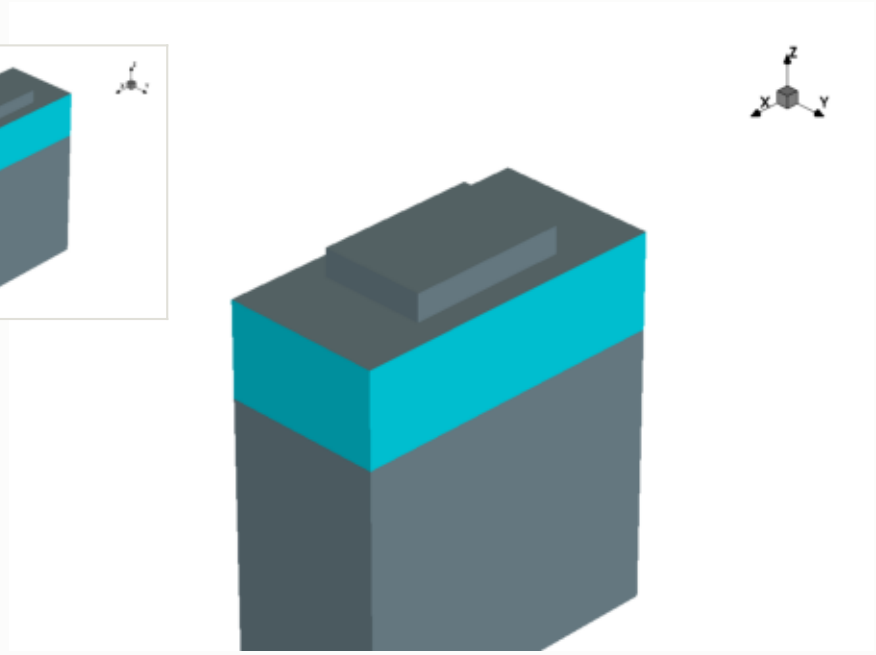
etch_strip · Wet_etch

0.169	89.1	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

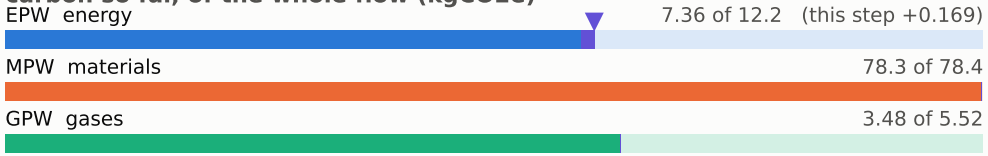




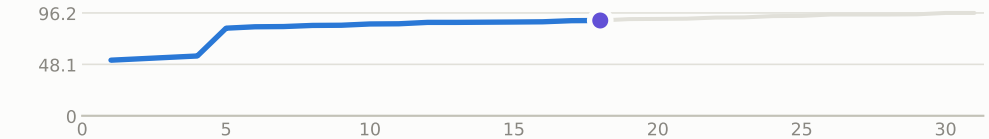
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

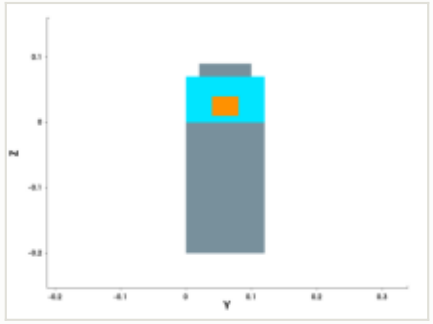
carbon so far, of the whole flow (kgCO2e)



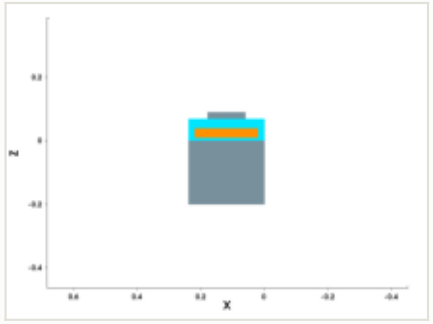
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



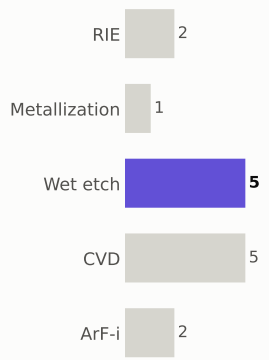
y cut at 0.060 µm



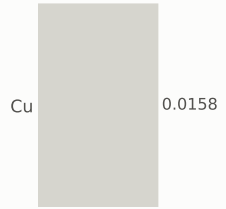
mask for this step

no mask for this step

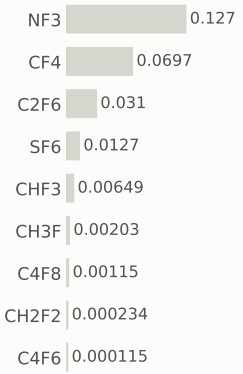
steps by process area



materials deposited (g)



gases emitted (g)



ILD

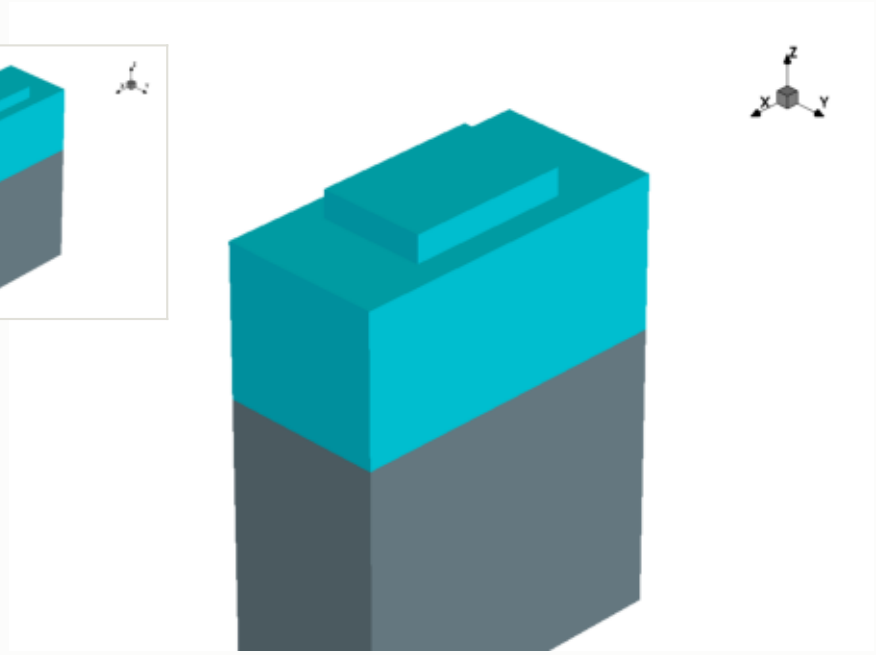
depo_anisotropic · CVD

1.31	90.4	1.4%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

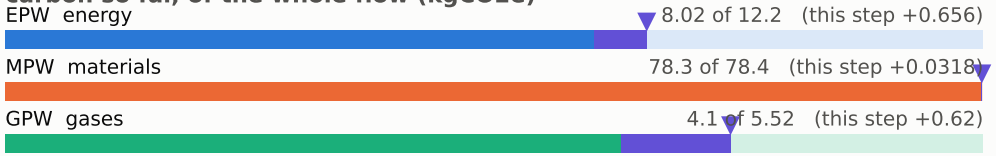




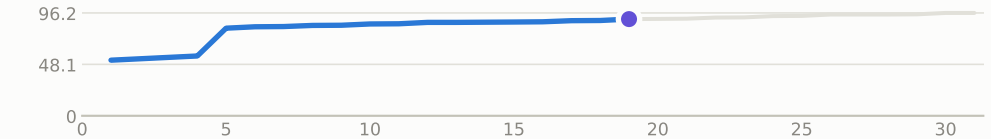
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.04 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

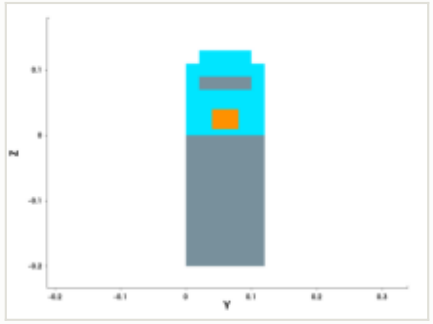
carbon so far, of the whole flow (kgCO2e)



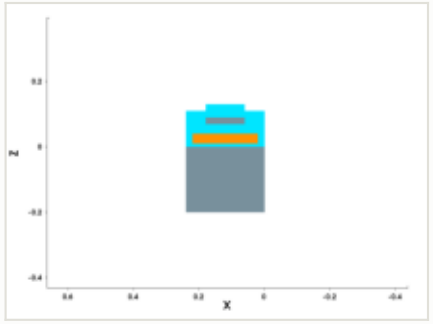
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



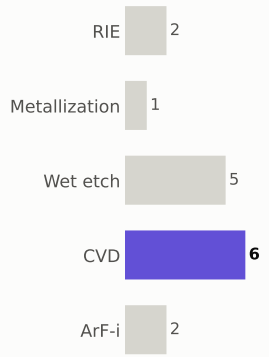
y cut at 0.060 µm



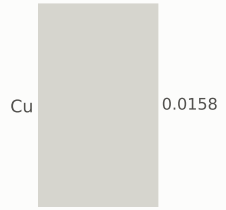
mask for this step

no mask for this step

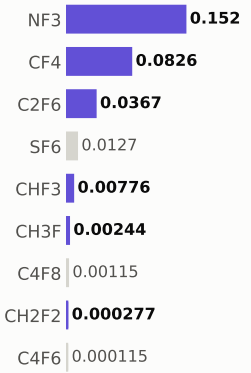
steps by process area



materials deposited (g)



gases emitted (g)



ILD CMP

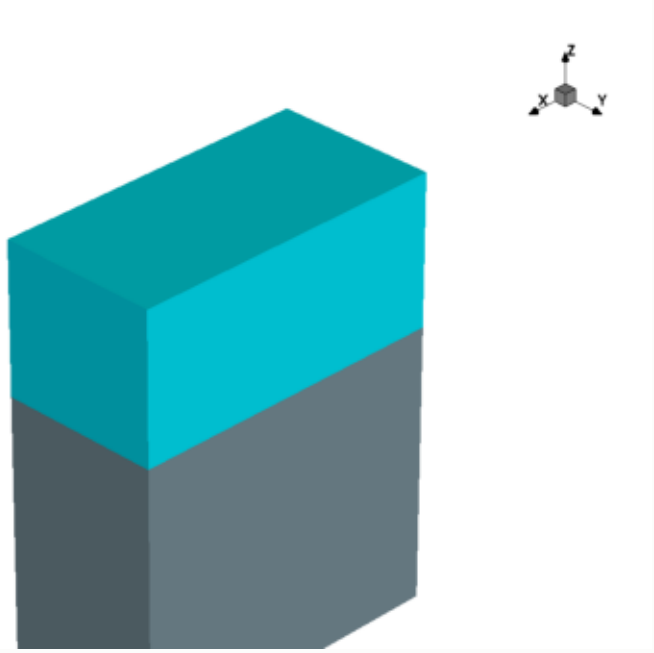
etch_cmp · Wet_etch

0.169	90.6	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

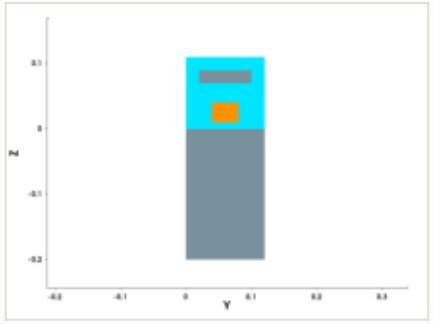
3D structure after this step

zoomed to fit

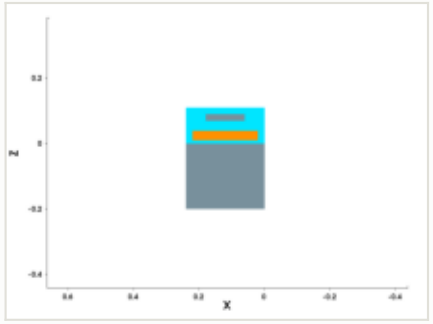




x cut at 0.120 µm



y cut at 0.060 µm



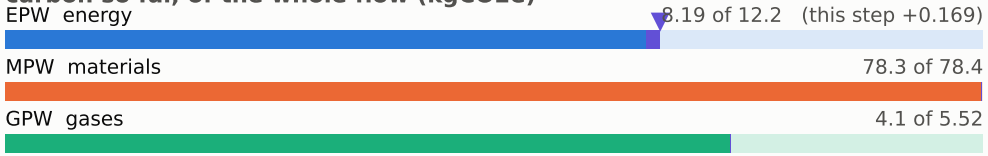
mask for this step

no mask for this step

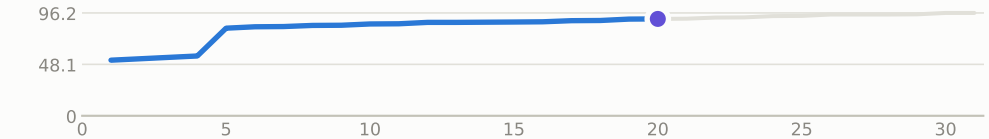
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

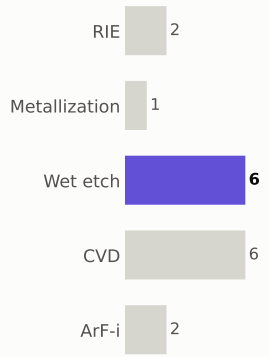
carbon so far, of the whole flow (kgCO2e)



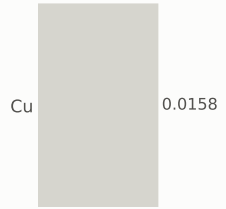
embodied carbon up to each step (kgCO2e per wafer)



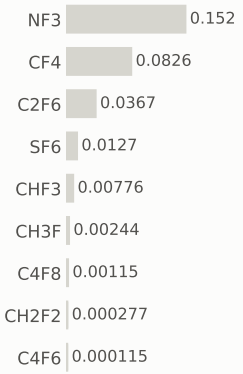
steps by process area



materials deposited (g)



gases emitted (g)



Top metal litho

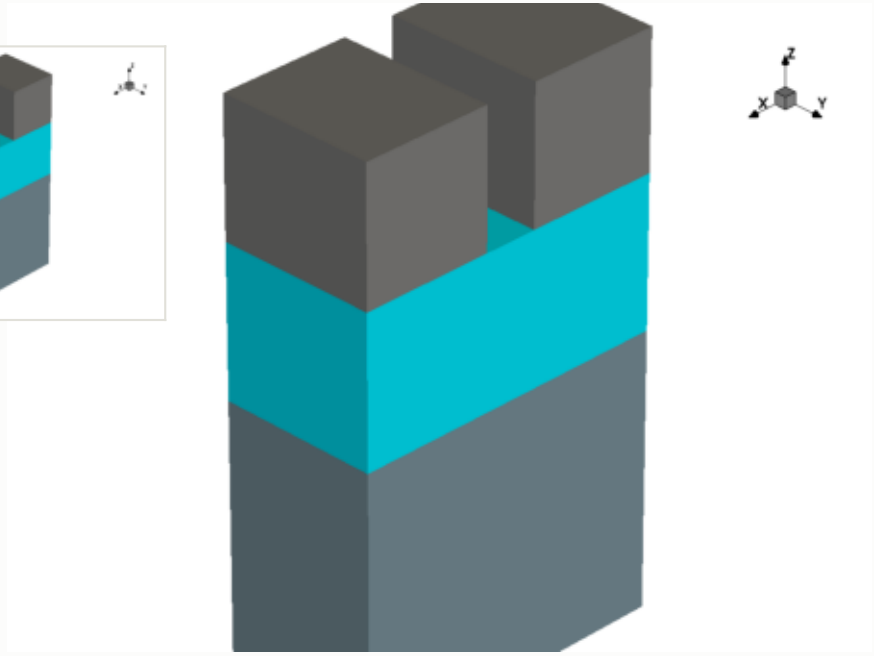
litho_patterning · ArF-i

0.217	90.8	0.23%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

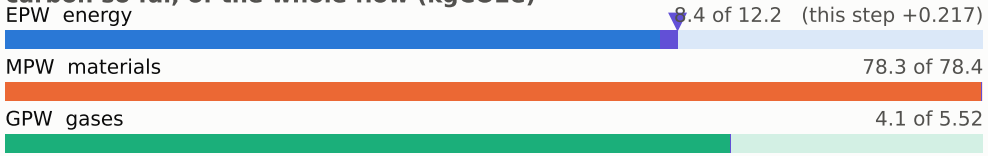




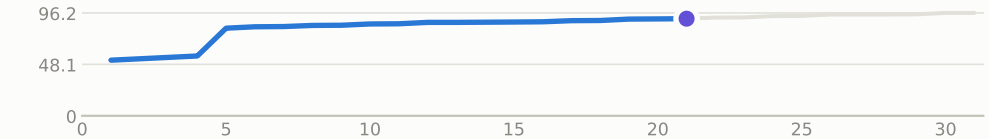
this step

process area	ArF-i	EPW	0.217 kgCO2e
material	—	EPW tool	0.155 kgCO2e
thickness	—	EPW overhead	0.0619 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.36 kWh + 0.144 overhead	embodied carbon	**0.217 kgCO2e**

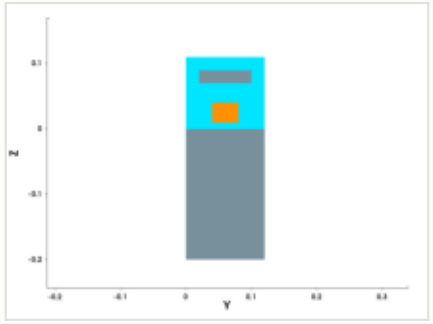
carbon so far, of the whole flow (kgCO2e)



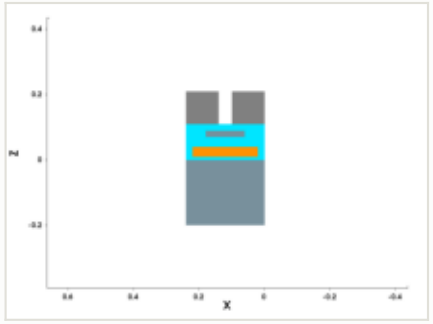
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



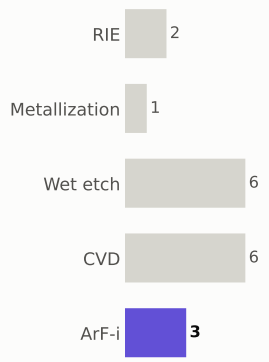
y cut at 0.060 µm



mask: layer 3:0 (GDSII)



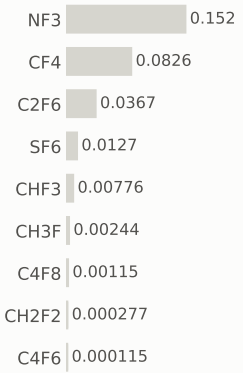
steps by process area



materials deposited (g)



gases emitted (g)



Top metal trench etch

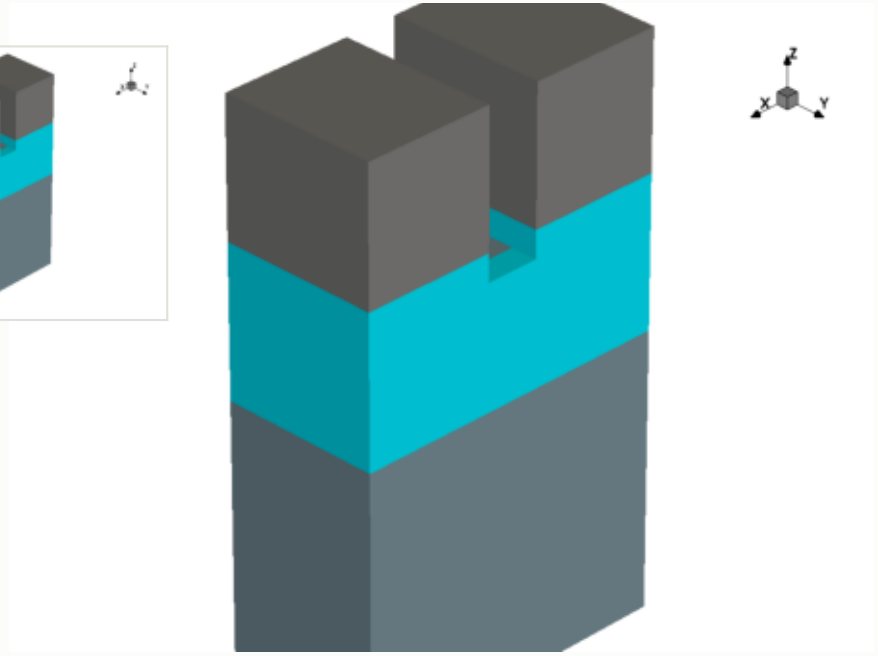
etch_anisotropic · RIE

1.04	91.9	1.1%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

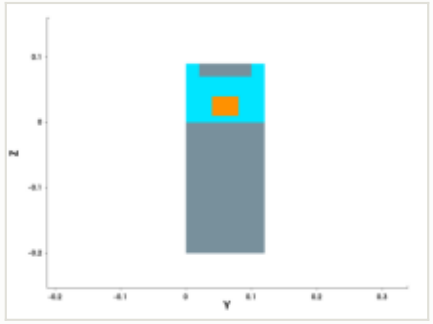
3D structure after this step

zoomed to fit

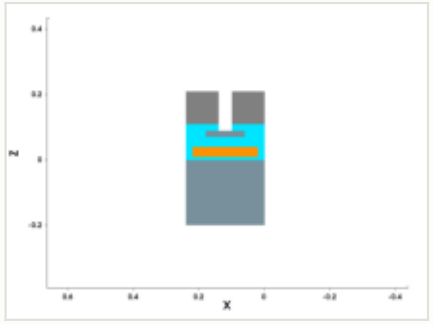




x cut at 0.120 µm



y cut at 0.060 µm



mask for this step

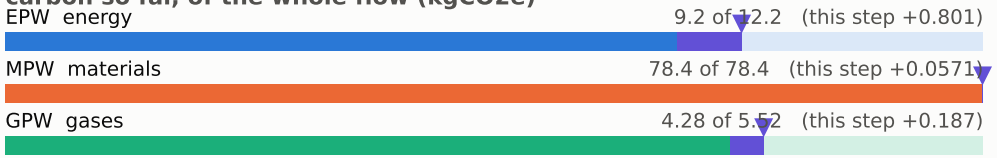
no mask for this step

this step

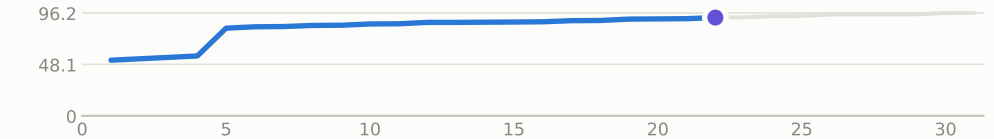
process area	RIE
material	—
thickness	—
material GWP	—
input gases	C4F8 0.16 g, SF6 0.439 g
energy	1.33 kWh + 0.532 overhead

EPW	0.801 kgCO2e
EPW tool	0.572 kgCO2e
EPW overhead	0.229 kgCO2e
MPW	0.0571 kgCO2e (gas 0.0571)
GPW	0.187 kgCO2e
embodied carbon	**1.04 kgCO2e**

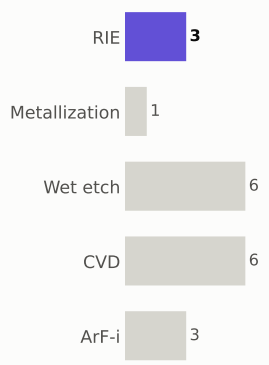
carbon so far, of the whole flow (kgCO2e)



embodied carbon up to each step (kgCO2e per wafer)



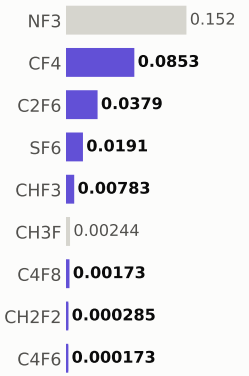
steps by process area



materials deposited (g)



gases emitted (g)



Resist strip

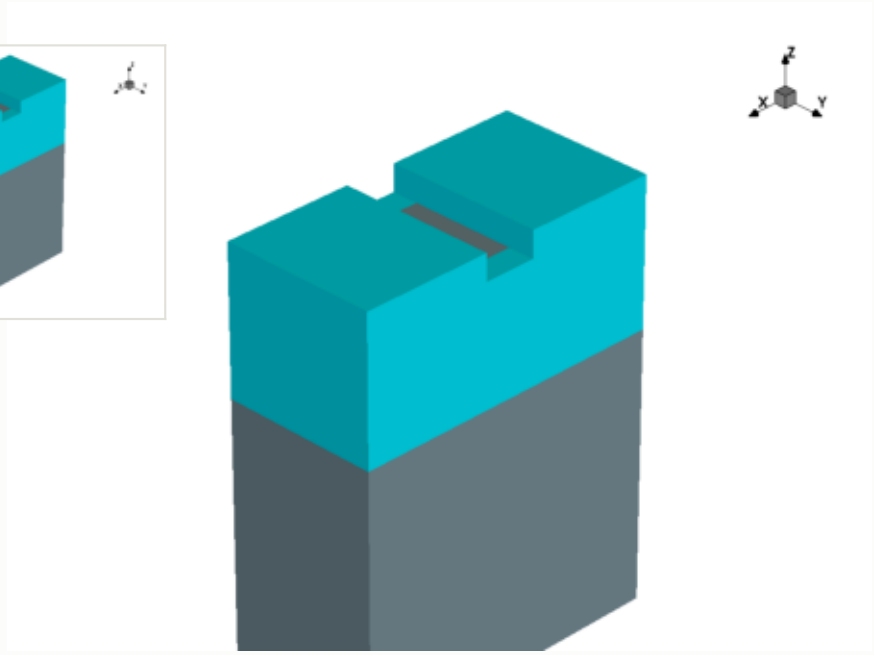
etch_strip · Wet_etch

0.169	92	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

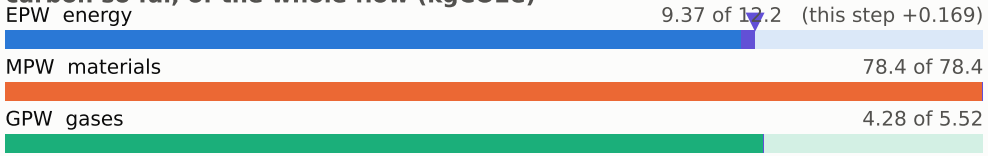




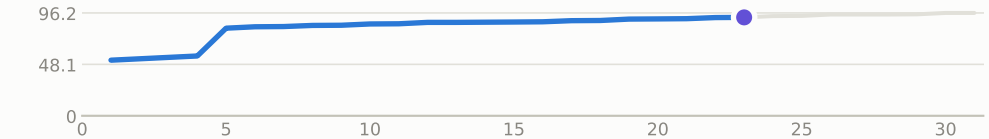
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

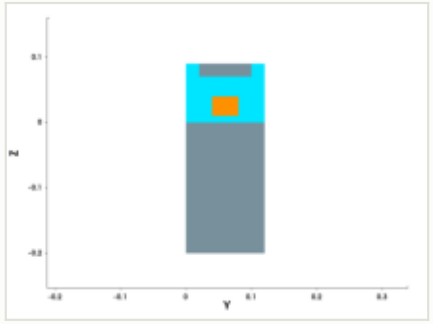
carbon so far, of the whole flow (kgCO2e)



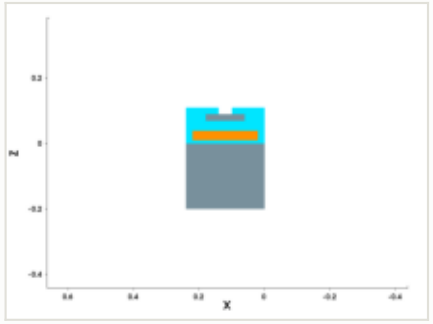
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



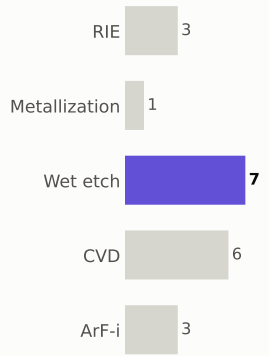
y cut at 0.060 µm



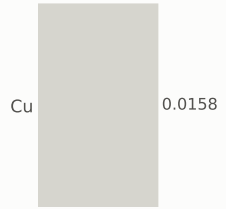
mask for this step

no mask for this step

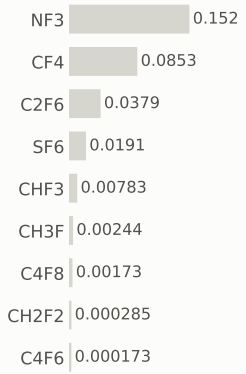
steps by process area



materials deposited (g)



gases emitted (g)



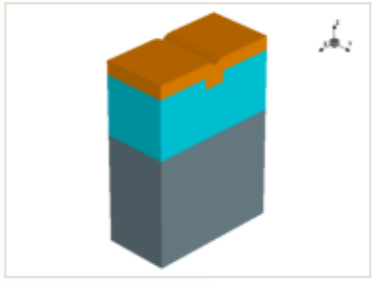
Cu fill

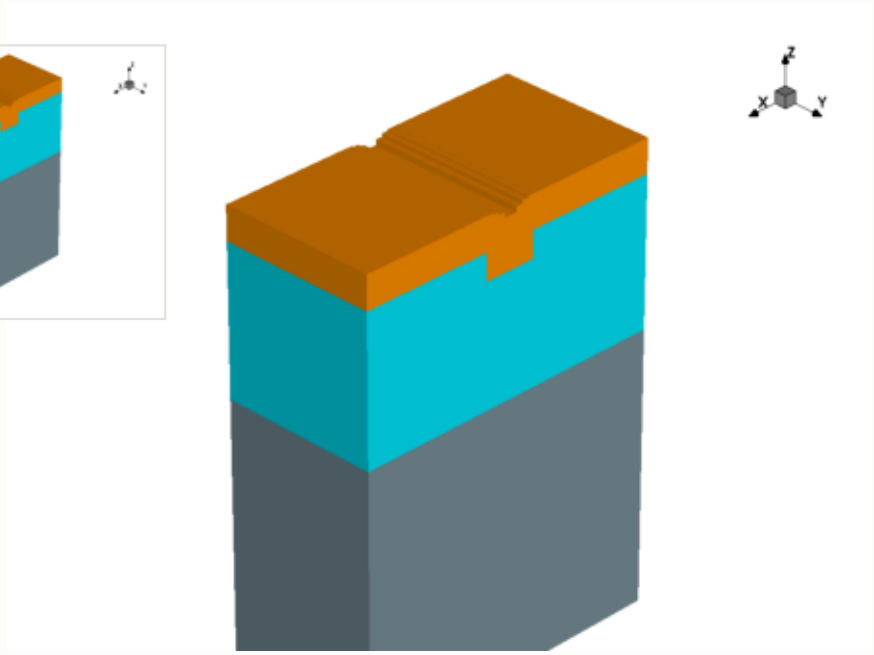
depo_isotropic · Metallization

1.2	93.2	1.3%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

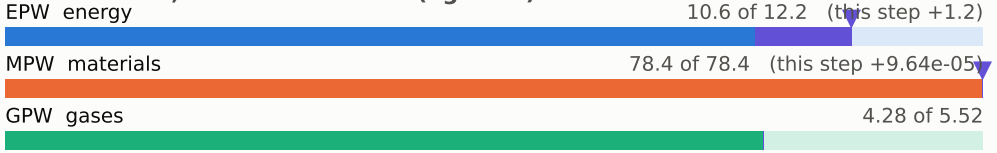




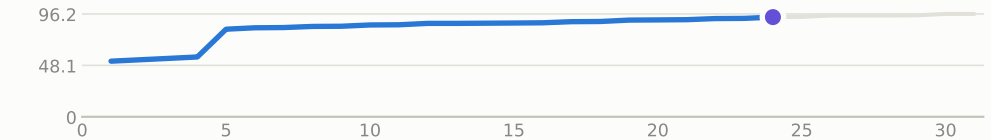
this step

process area	Metallization	EPW	1.2 kgCO2e
material	Cu	EPW tool	0.86 kgCO2e
thickness	0.025 µm	EPW overhead	0.344 kgCO2e
material GWP	6.09011 gCO2e/g	MPW	9.64e-05 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	2 kWh + 0.8 overhead	embodied carbon	**1.2 kgCO2e**

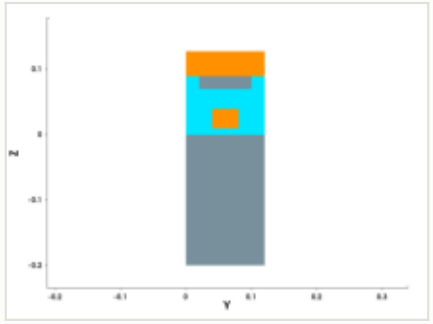
carbon so far, of the whole flow (kgCO2e)



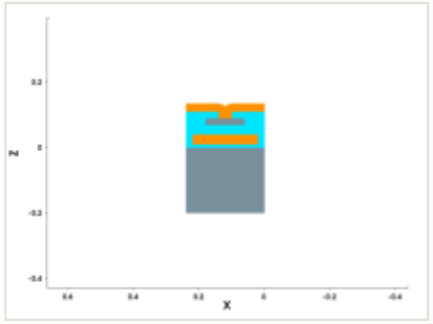
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



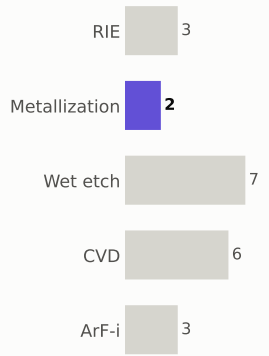
y cut at 0.060 µm



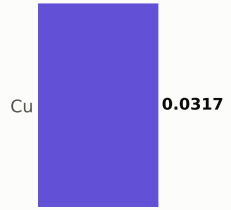
mask for this step

no mask for this step

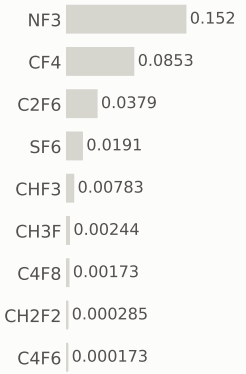
steps by process area



materials deposited (g)



gases emitted (g)



Cu CMP

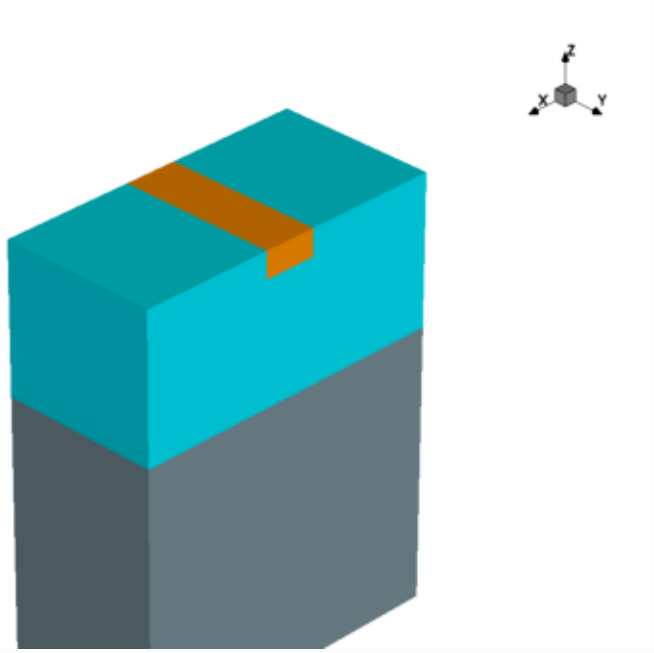
etch_cmp · Wet_etch

0.169	93.4	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

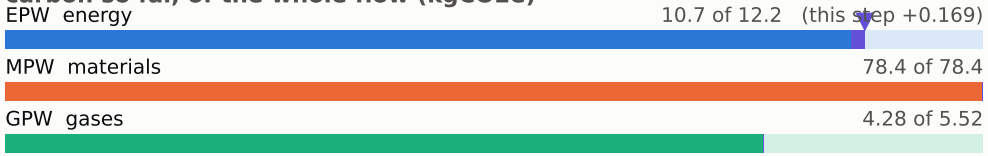




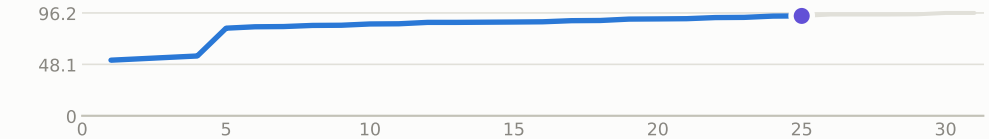
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

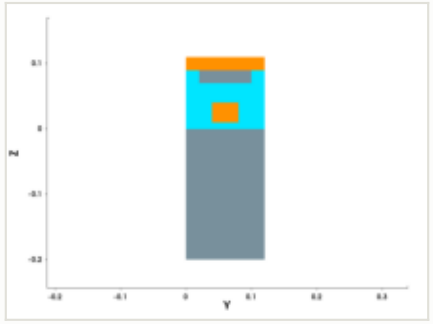
carbon so far, of the whole flow (kgCO2e)



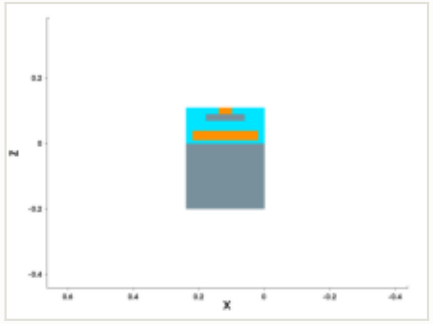
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



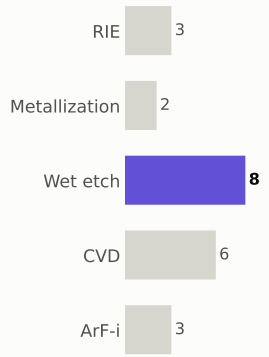
y cut at 0.060 µm



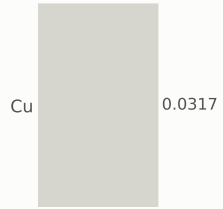
mask for this step

no mask for this step

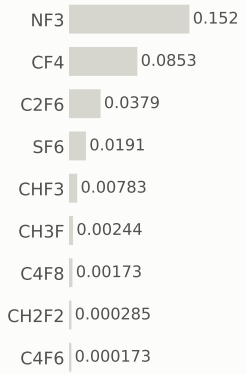
steps by process area



materials deposited (g)



gases emitted (g)



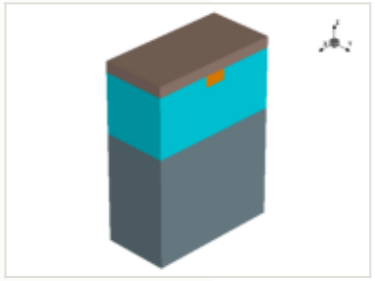
Bonding dielectric

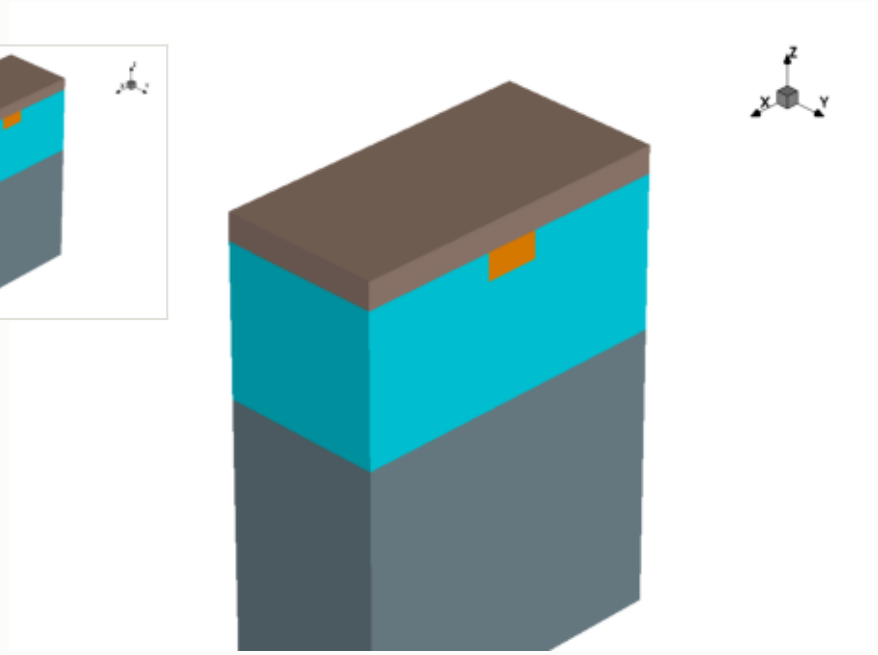
depo_anisotropic · CVD

1.31	94.7	1.4%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

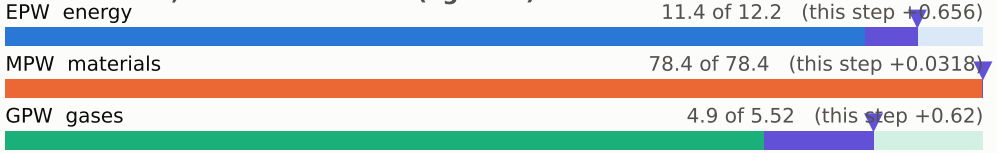




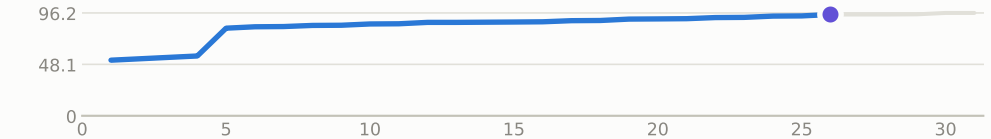
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.02 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

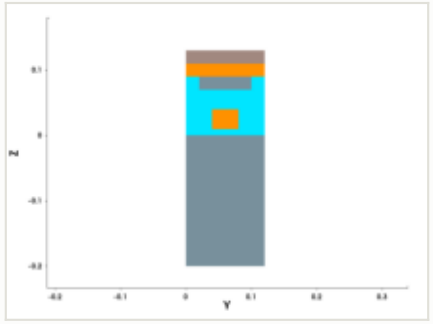
carbon so far, of the whole flow (kgCO2e)



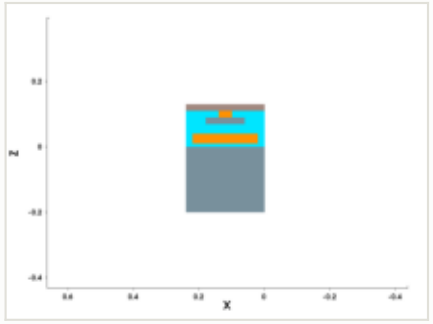
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



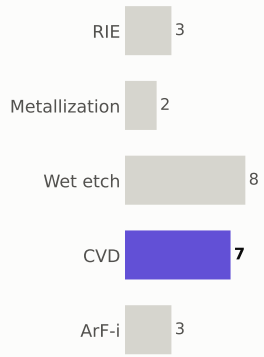
y cut at 0.060 µm



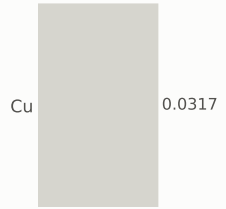
mask for this step

no mask for this step

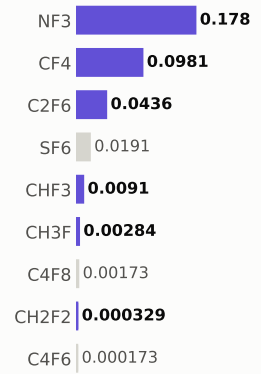
steps by process area



materials deposited (g)



gases emitted (g)



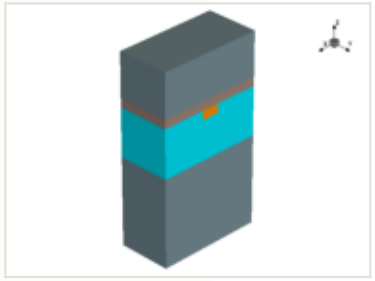
Bond carrier wafer

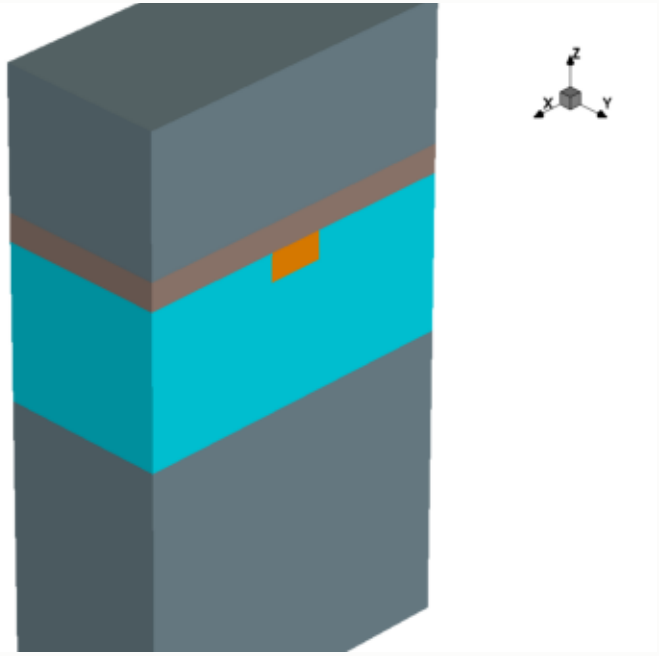
bond_structure · Wafer

0	94.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

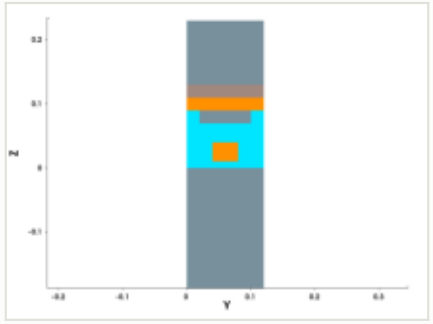
3D structure after this step

zoomed to fit

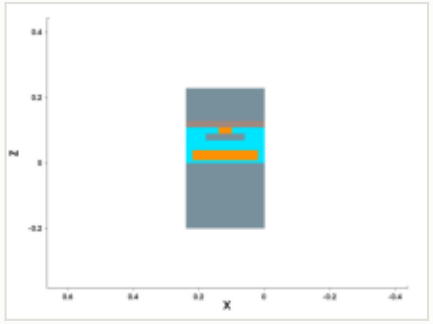




x cut at 0.120 µm



y cut at 0.060 µm



mask for this step

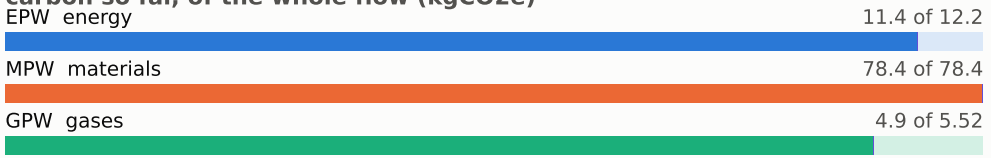
no mask for this step

this step

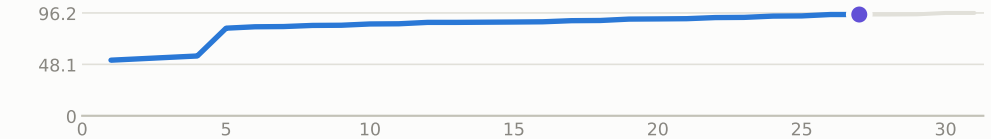
process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

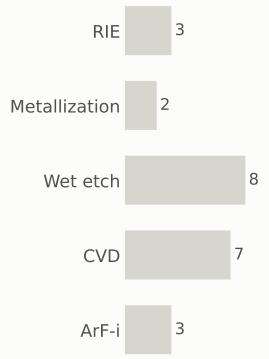
carbon so far, of the whole flow (kgCO2e)



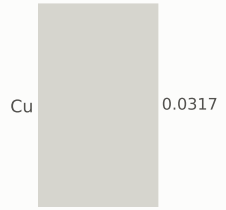
embodied carbon up to each step (kgCO2e per wafer)



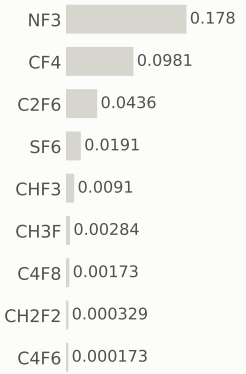
steps by process area



materials deposited (g)



gases emitted (g)



Flip stack

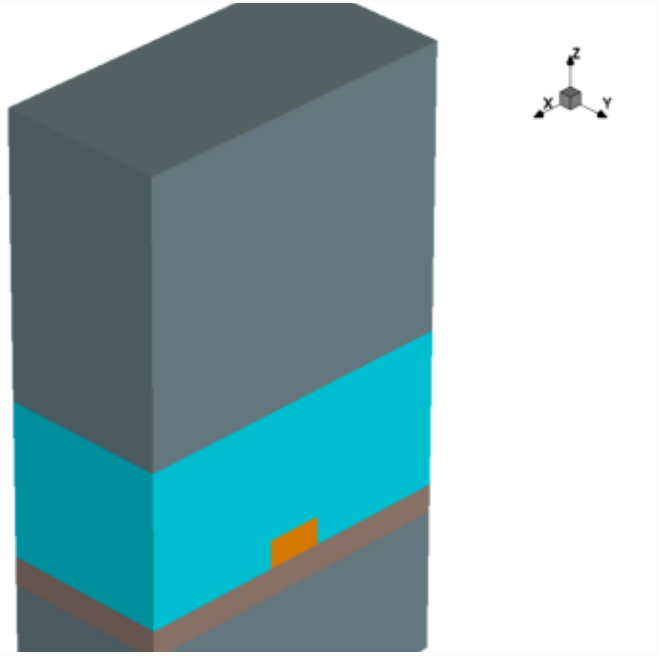
flip_structure · Wafer

0	94.7	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



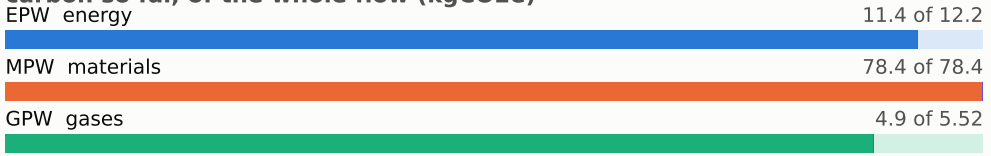


this step

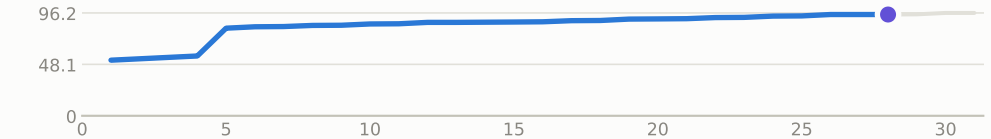
process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

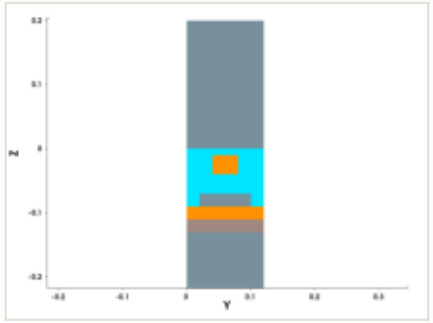
carbon so far, of the whole flow (kgCO2e)



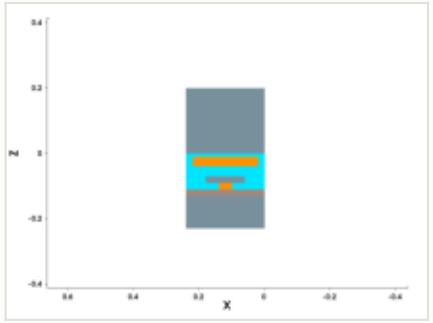
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



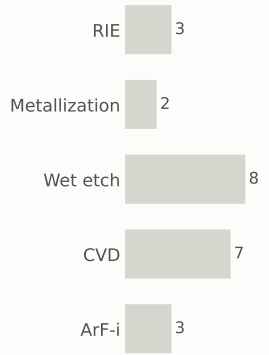
y cut at 0.060 µm



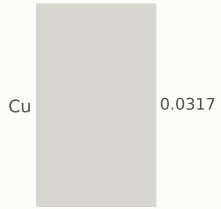
mask for this step

no mask for this step

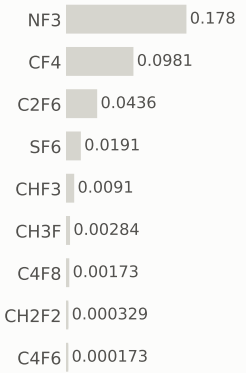
steps by process area



materials deposited (g)



gases emitted (g)



Backside thinning CMP

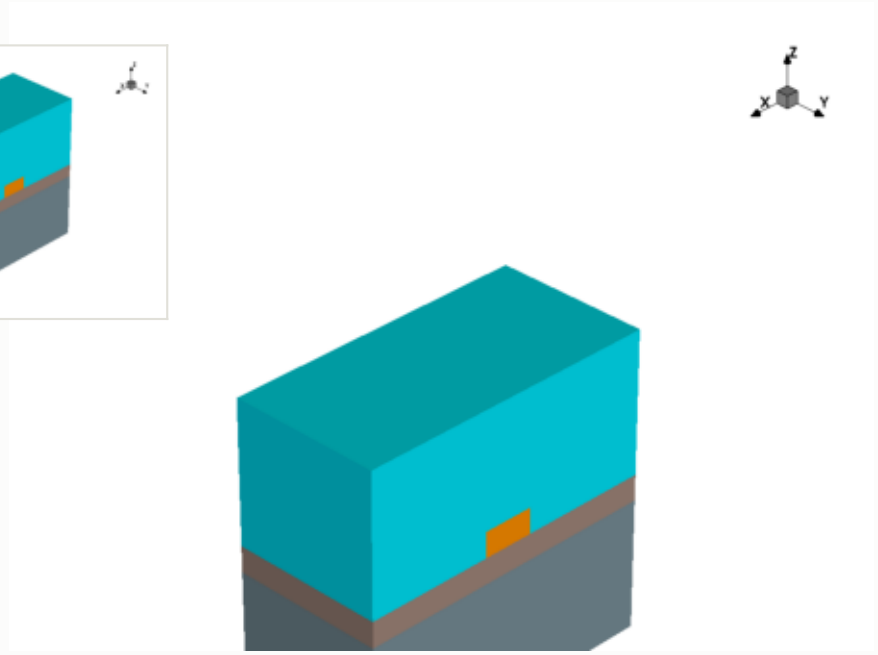
etch_cmp · Wet_etch

0.169	94.9	0.18%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

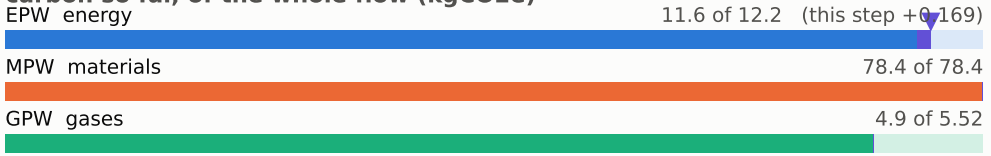




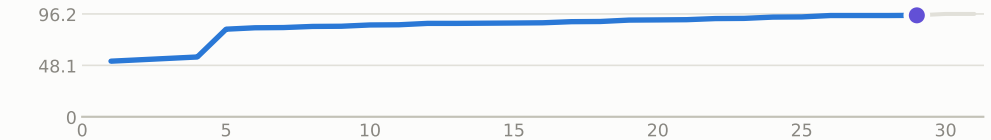
this step

process area	Wet_etch	EPW	0.169 kgCO2e
material	—	EPW tool	0.12 kgCO2e
thickness	—	EPW overhead	0.0482 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0.28 kWh + 0.112 overhead	embodied carbon	**0.169 kgCO2e**

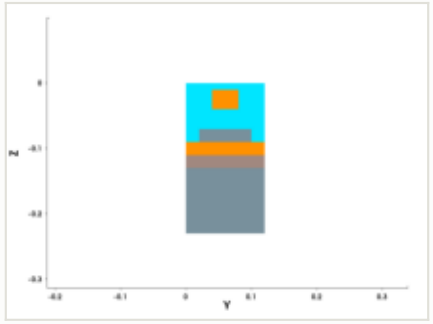
carbon so far, of the whole flow (kgCO2e)



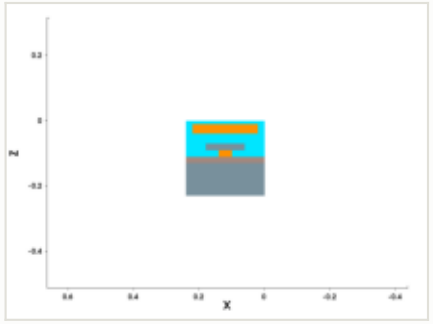
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



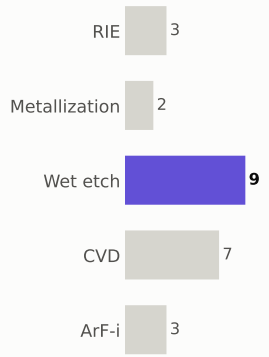
y cut at 0.060 µm



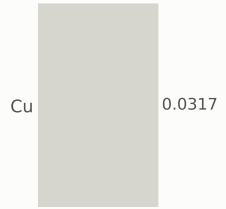
mask for this step

no mask for this step

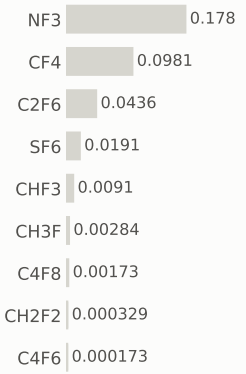
steps by process area



materials deposited (g)



gases emitted (g)



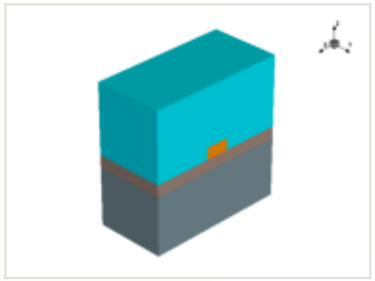
Backside passivation

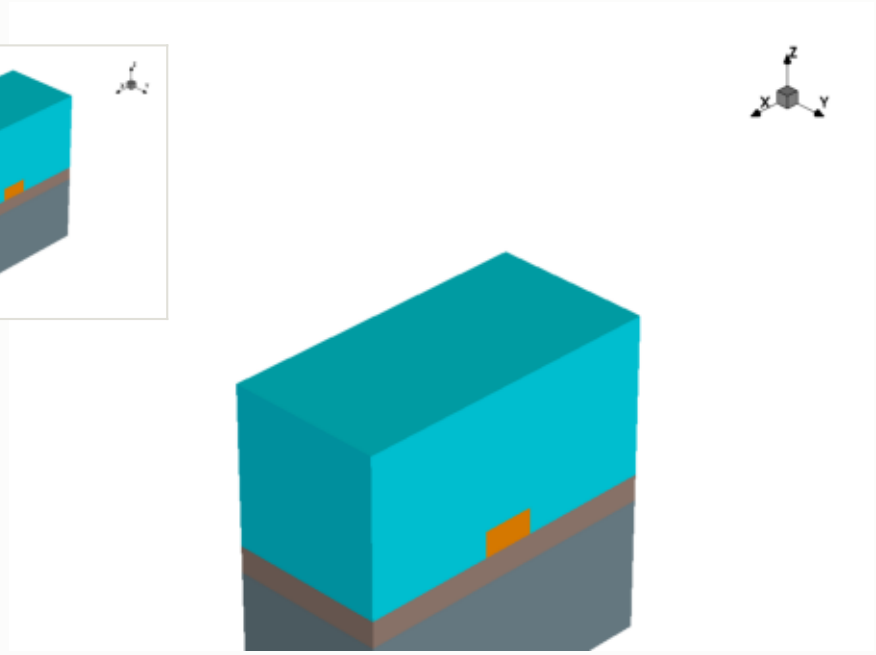
depo_anisotropic · CVD

1.31	96.2	1.4%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit

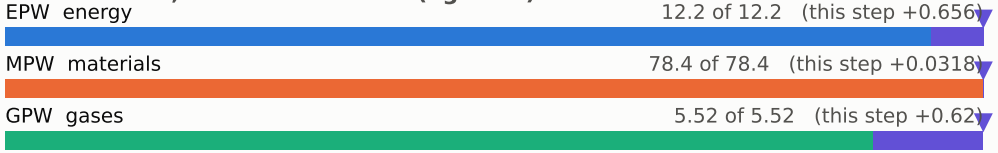




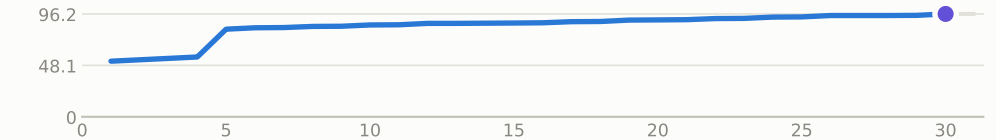
this step

process area	CVD	EPW	0.656 kgCO2e
material	—	EPW tool	0.469 kgCO2e
thickness	0.01 µm	EPW overhead	0.187 kgCO2e
material GWP	—	MPW	0.0318 kgCO2e (gas 0.0318)
input gases	NF3 2.54 g	GPW	0.62 kgCO2e
energy	1.09 kWh + 0.436 overhead	embodied carbon	**1.31 kgCO2e**

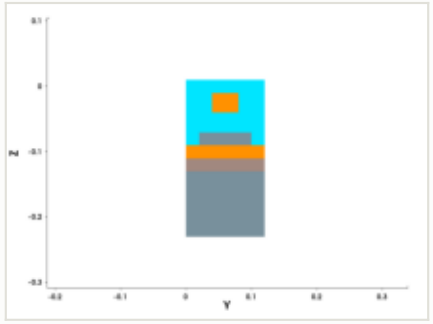
carbon so far, of the whole flow (kgCO2e)



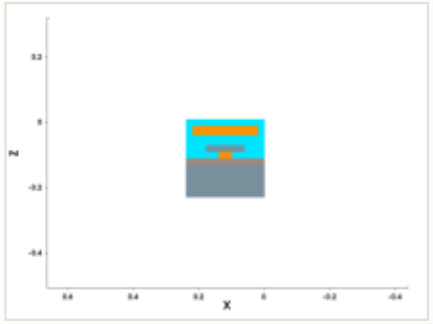
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



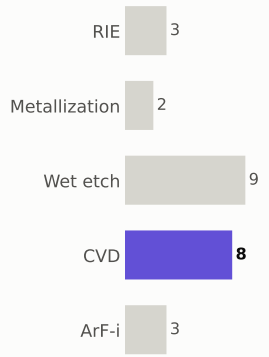
y cut at 0.060 µm



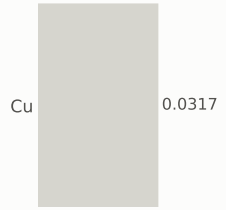
mask for this step

no mask for this step

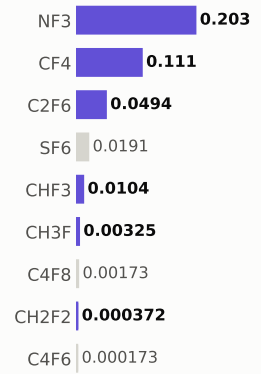
steps by process area



materials deposited (g)



gases emitted (g)



Flip stack

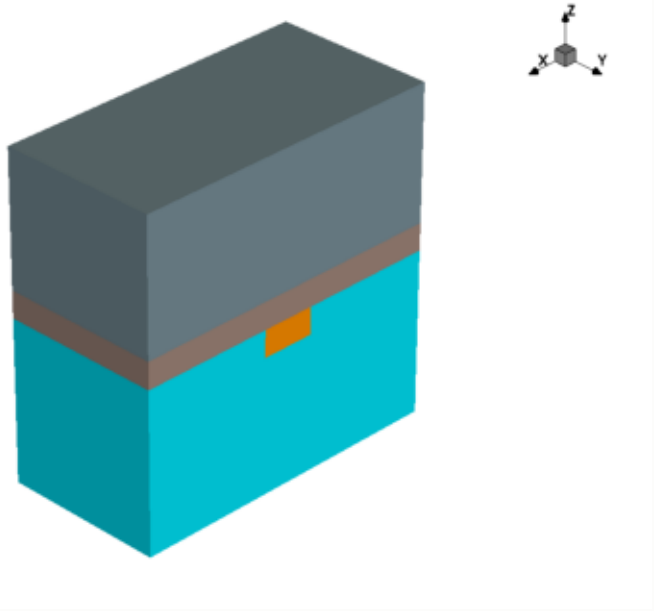
flip_structure · Wafer

0	96.2	0%
this step (kgCO2e)	flow so far (kgCO2e)	share of the flow

3D structure after this step

zoomed to fit



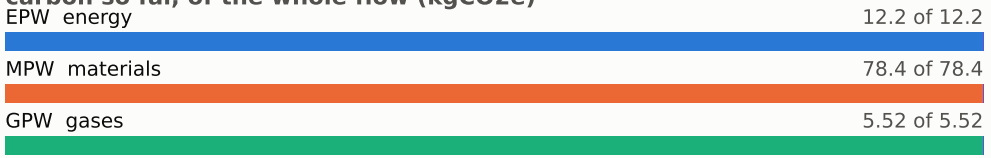


this step

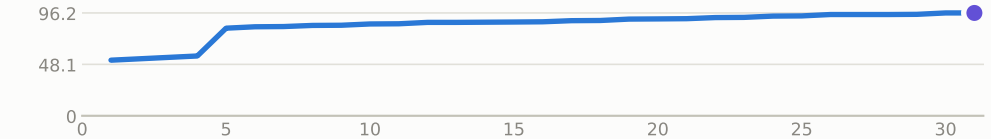
process area	Wafer	EPW	0 kgCO2e
material	—	EPW tool	0 kgCO2e
thickness	—	EPW overhead	0 kgCO2e
material GWP	—	MPW	0 kgCO2e
input gases	—	GPW	0 kgCO2e
energy	0 kWh + 0 overhead	embodied carbon	**0 kgCO2e**

Wafer step: not counted as a process step (uncounted_process_areas)

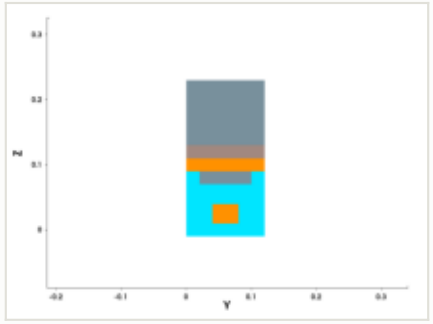
carbon so far, of the whole flow (kgCO2e)



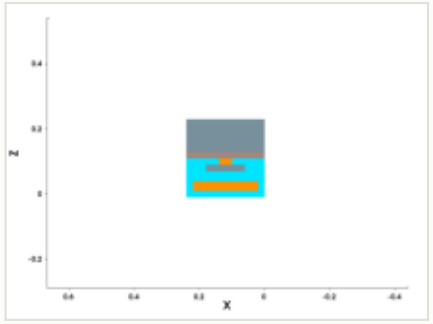
embodied carbon up to each step (kgCO2e per wafer)



x cut at 0.120 µm



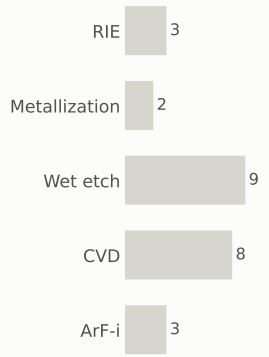
y cut at 0.060 µm



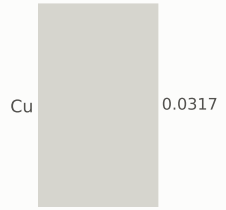
mask for this step

no mask for this step

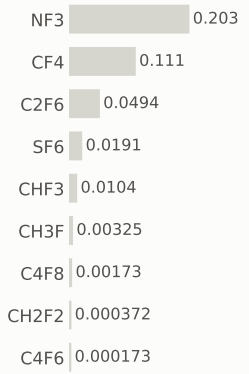
steps by process area



materials deposited (g)

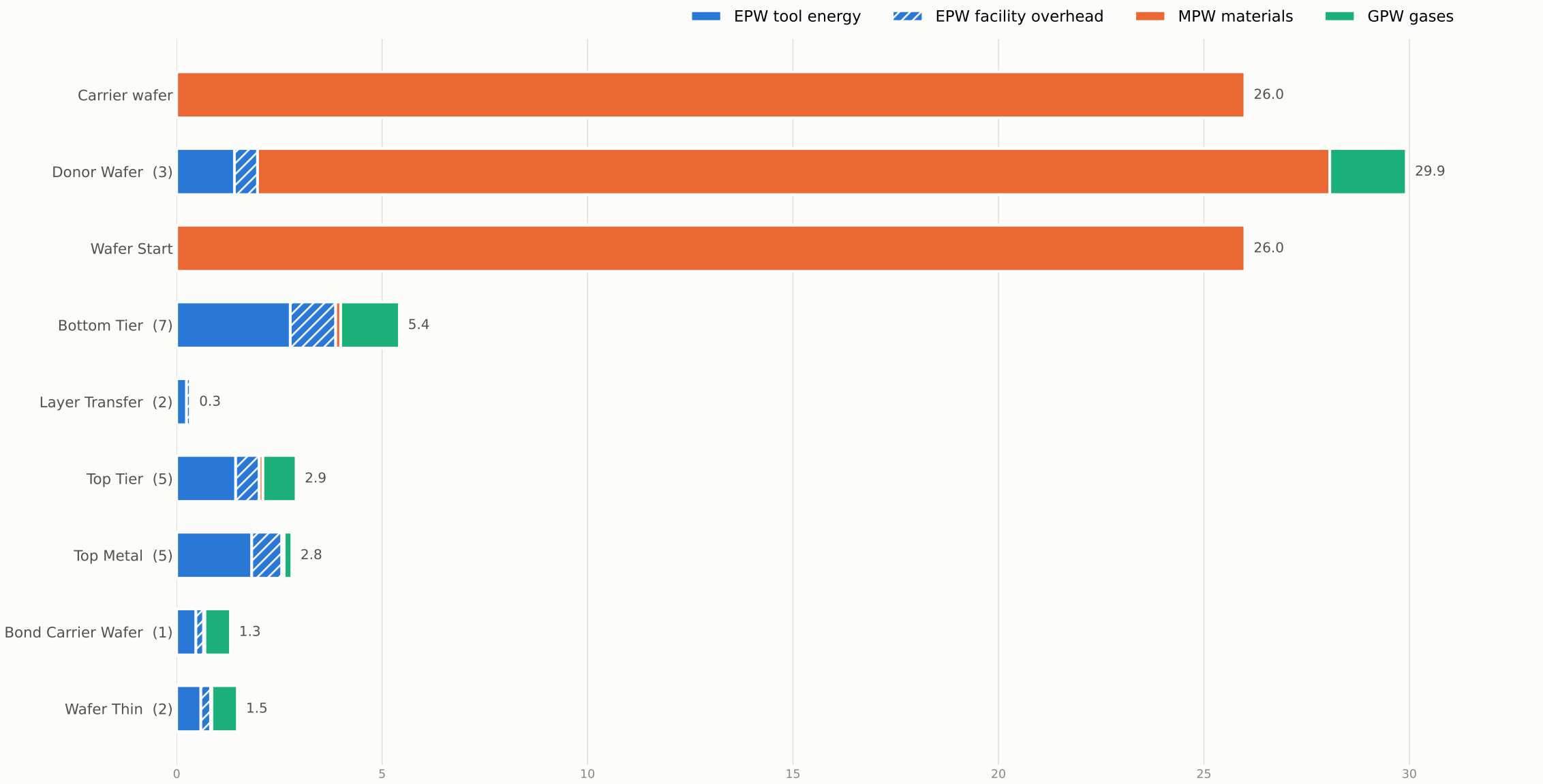


gases emitted (g)



Carbon by module

kgCO2e per wafer, in flow order (steps in brackets)



Totals by process area

kgCO2e per wafer; EPW = tool + facility overhead (overhead energy = tool energy x 0.4)

process area	steps	tool kWh	overhead kWh	EPW tool	EPW overhead	EPW	MPW	GPW	total
CVD	8	8.72	3.49	3.75	1.50	5.25	0.25	4.96	10.47
RIE	3	3.99	1.60	1.72	0.69	2.40	0.17	0.56	3.13
Metallization	2	4.00	1.60	1.72	0.69	2.41	0.00	0.00	2.41
Wet_etch	9	2.52	1.01	1.08	0.43	1.52	0.00	0.00	1.52
ArF-i	3	1.08	0.43	0.46	0.19	0.65	0.00	0.00	0.65
Wafer		0.00	0.00	0.00	0.00	0.00	78.00	0.00	78.00
all	25	20.31	8.12	8.73	3.49	12.23	78.43	5.52	96.18